Quantitative

An open, game-like syllabus for the quantitative skills behind top STEM careers

72 topics · 758 lectures · about 2274 hours of instruction

Seven domains: Math, Probability & Statistics, AI / ML, Derivatives & Risk, Markets & Trading, Problem Solving, Computing. Every topic lists its prerequisites, matching university courses, a lecture-by-lecture outline and what you should be able to do at the end.

Course codes and lecture structures are modelled on the Hong Kong university courses cited in each topic. Some codes are unverified and are flagged in the notes. Edition: October 2026 draft.

How this syllabus works

Three parts per topic. Every topic has **Theory** (multiple-choice concept questions), **Computation** (numeric answer-only questions with parametrised numbers) and **Proof** (written arguments, graded later with rubric-based AI marking).

Sufficient and Complete. Finishing Theory and Computation makes a topic **Sufficient**. Adding Proof makes it **Complete**. Sufficient is enough to unlock the next topic and to count towards the required topics of a career path.

Coding. Coding is an input mode of Computation, not a fourth part. Programming topics use LeetCode-style hidden test cases; all tests must pass. Stochastic programs are graded with a fixed seed and a tolerance. Python and SQL run in the browser first; C++ comes later.

Topics with no natural proofs (Mental Math, Fermi Estimation, Dev Tools, LaTeX, Python, C++, MLOps and similar) have Proof = N/A. Complete means passing the hardest question set.

Retakes. Unlimited, with newly parametrised questions weighted towards the lectures where you were weak. A placement test lets you skip topics you already know.

Course format. Each topic is 8-12 lectures of 3 hours. Where a university course matches, the outline follows it closely; lectures are added where it falls short. Each lecture lists Theory items and Computation items, plus a Coding task where Python is a prerequisite.

Suggested pass marks (to be confirmed): Theory 80%, Computation 75%.

Course index

Code	Title	Domain	Stage	Lectures	Prerequisites
QNT-101	Calculus	Math	1	11	None
QNT-102	Discrete Math & Combinatorics	Math	1	10	None
QNT-103	Linear Algebra I	Math	1	10	None
QNT-104	Differential Equations	Math	2	11	QNT-101, QNT-103
QNT-105	Linear Algebra II	Math	2	10	QNT-103
QNT-106	Multivariable Calculus	Math	2	10	QNT-101
QNT-107	Numerical Methods I	Math	2	10	QNT-101, QNT-103, QNT-705
QNT-108	Real Analysis	Math	2	10	QNT-101
QNT-109	Numerical Methods II	Math	3	10	QNT-107, QNT-104
QNT-110	Optimization	Math	3	10	QNT-106, QNT-103
QNT-111	Information Theory	Math	4	11	QNT-202
QNT-201	Probability I	Probability & Statistics	2	12	QNT-101
QNT-202	Probability II	Probability & Statistics	3	11	QNT-201, QNT-106
QNT-203	Regression I	Probability & Statistics	3	11	QNT-201, QNT-103
QNT-204	Monte Carlo	Probability & Statistics	4	11	QNT-202, QNT-705
QNT-205	Probability III	Probability & Statistics	4	11	QNT-202, QNT-108
QNT-206	Statistical Inference	Probability & Statistics	4	11	QNT-202
QNT-207	Stochastic Processes	Probability & Statistics	4	11	QNT-202
QNT-208	Bayesian	Probability & Statistics	5	12	QNT-206, QNT-705
QNT-209	Causal Inference	Probability & Statistics	5	10	QNT-203, QNT-206
QNT-210	Experimental Design / A/B Testing	Probability & Statistics	5	11	QNT-206, QNT-203
QNT-211	Nonparametric	Probability & Statistics	5	10	QNT-206
QNT-212	Regression II	Probability & Statistics	5	11	QNT-203, QNT-206, QNT-105
QNT-213	Research Methods	Probability & Statistics	5	11	QNT-203, QNT-301
QNT-214	Time Series I	Probability & Statistics	5	10	QNT-203, QNT-207
QNT-215	High-dimensional Stats	Probability & Statistics	6	10	QNT-212, QNT-105
QNT-216	Time Series II	Probability & Statistics	6	10	QNT-214, QNT-206
QNT-301	Machine Learning	AI / ML	4	11	QNT-203, QNT-110, QNT-705
QNT-302	Deep Learning	AI / ML	5	11	QNT-301
QNT-303	Learning Theory	AI / ML	5	10	QNT-301, QNT-205
QNT-304	MLOps	AI / ML	5	10	QNT-301, QNT-710, QNT-703
QNT-305	Reinforcement Learning	AI / ML	5	11	QNT-301, QNT-207
QNT-306	Computer Vision	AI / ML	6	10	QNT-302

Code	Title	Domain	Stage	Lectures	Prerequisites
QNT-307	NLP	AI / ML	6	10	QNT-302
QNT-308	LLM / Generative AI	AI / ML	7	11	QNT-307
QNT-401	Derivatives I	Derivatives & Risk	3	12	QNT-201
QNT-402	Credit Risk	Derivatives & Risk	4	10	QNT-401, QNT-202
QNT-403	Interest Rate I	Derivatives & Risk	4	10	QNT-401
QNT-404	Risk Management / VaR	Derivatives & Risk	4	11	QNT-502, QNT-202
QNT-405	Risk Theory & Insurance	Derivatives & Risk	4	10	QNT-202
QNT-406	Operational Risk & Regulation	Derivatives & Risk	5	10	QNT-404
QNT-407	Stochastic Calculus	Derivatives & Risk	5	10	QNT-207, QNT-106
QNT-408	Derivatives II	Derivatives & Risk	6	10	QNT-401, QNT-407
QNT-409	Interest Rate II	Derivatives & Risk	6	10	QNT-403, QNT-407
QNT-410	Structured Products	Derivatives & Risk	7	10	QNT-408, QNT-204
QNT-411	Volatility Models	Derivatives & Risk	7	10	QNT-408, QNT-107
QNT-501	Microstructure I	Markets & Trading	3	10	QNT-201
QNT-502	Portfolio & Asset Pricing I	Markets & Trading	3	10	QNT-201, QNT-106
QNT-503	Market Making	Markets & Trading	4	10	QNT-501, QNT-603, QNT-604
QNT-504	Options & Vol Trading	Markets & Trading	4	11	QNT-401, QNT-603
QNT-505	Portfolio & Asset Pricing II	Markets & Trading	4	10	QNT-502, QNT-110, QNT-203
QNT-506	Microstructure II	Markets & Trading	5	10	QNT-501, QNT-207, QNT-110
QNT-507	Algo / Quant Trading	Markets & Trading	6	11	QNT-203, QNT-214, QNT-705, QNT-501
QNT-601	Fermi Estimation	Problem Solving	1	10	None
QNT-602	Mental Math	Problem Solving	1	10	None
QNT-603	EV & Kelly	Problem Solving	3	10	QNT-201
QNT-604	Game Theory	Problem Solving	3	10	QNT-201
QNT-605	Probability Puzzles	Problem Solving	3	10	QNT-201, QNT-102
QNT-701	C++ I	Computing	1	11	None
QNT-702	Computer Architecture	Computing	1	12	None
QNT-703	Dev Tools (Linux · Git)	Computing	1	10	None
QNT-704	LaTeX & Research Writing	Computing	1	10	None
QNT-705	Python	Computing	1	11	None
QNT-706	DSA I	Computing	2	12	QNT-705, QNT-102
QNT-707	C++ II	Computing	3	10	QNT-701, QNT-706
QNT-708	Computer Networks	Computing	3	11	QNT-706
QNT-709	DSA II	Computing	3	11	QNT-706
QNT-710	Databases	Computing	3	11	QNT-706
QNT-711	Operating Systems	Computing	3	11	QNT-701, QNT-706, QNT-702
QNT-712	Distributed Systems	Computing	4	11	QNT-711, QNT-708
QNT-713	Low-latency Programming	Computing	4	11	QNT-707, QNT-711
QNT-714	Software Engineering & System Design	Computing	4	11	QNT-706, QNT-710

QNT-101 · Math

Calculus

Stage 1
11 lectures x 3 h

Single-variable calculus: limits, derivatives, integrals and series.

Prerequisites	None
University courses	HKU MATH1013 · CUHK MATH1010 · CityU MA1200/1201
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Differentiate and integrate standard functions
- Evaluate limits and series
- Apply calculus to optimisation and approximation

Course outline

#	Topic	Theory	Computation / Coding
1	**Sequences and limits**	Convergence, limit laws, squeeze theorem, monotone convergence	Limits of sequences
2	**Limits of functions**	One-sided limits, limits at infinity, standard limits	Evaluating limits
3	**Continuity**	Definition, types of discontinuity, intermediate value theorem	Points of continuity, bracketing a root
4	**Derivatives**	Definition, product, quotient and chain rules, implicit differentiation	Derivative values, tangent lines
5	**Applications of derivatives**	Curve sketching, optimization, related rates	Optimal values, rates of change
6	**Mean value theorem**	Rolle's theorem, the mean value theorem, monotonicity and concavity	The point c in the mean value theorem, intervals of increase
7	**L'Hôpital's rule and Taylor series**	Indeterminate forms, Taylor polynomials, remainder terms	Limits by L'Hôpital, Taylor approximations and error bounds
8	**Methods of integration**	Substitution, integration by parts, partial fractions, trigonometric substitution	Definite integrals by technique
9	**Integration and the fundamental theorem**	Riemann sums, the definite integral, both parts of the fundamental theorem	Areas, derivatives of integral functions
10	**Improper integrals and applications**	Convergence of improper integrals, area, volume, average value	Improper integral values, volumes of revolution
11	**Infinite series**	Convergence tests, power series, radius of convergence	Sums of series, radius of convergence

Notes. Follows the topic order of the CUHK MATH1010 tutorials, from sequences to the fundamental theorem of calculus. Applications of derivatives, improper integrals and infinite series are added because that page does not list them.

Proof. Proofs will be added later.

QNT-102 · Math

Discrete Math & Combinatorics

Stage 1
10 lectures x 3 h

Logic, sets, counting, graphs and recurrences.

Prerequisites	None
University courses	HKU COMP2121 · HKUST COMP2711
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Prove simple statements by induction and contradiction
- Count with permutations, combinations and generating functions
- Solve recurrences and graph problems

Course outline

#	Topic	Theory	Computation / Coding
1	**Propositional logic and quantifiers**	Connectives, truth tables, quantifiers, negation	Truth tables, logical equivalences
2	**Proof techniques**	Direct proof, contradiction, induction, pigeonhole principle	Induction-based sums, pigeonhole bounds
3	**Sets, relations and functions**	Set operations, relations, equivalence relations, injective and surjective functions	Counting functions and relations
4	**Enumeration I**	Sum and product rules, permutations, combinations	Counting arrangements and selections
5	**Enumeration II**	Binomial theorem, inclusion-exclusion, distributions into boxes	Inclusion-exclusion counts, binomial coefficients
6	**Basic number theory**	Divisibility, gcd and the Euclidean algorithm, modular arithmetic	gcd, modular inverses, congruences
7	**Recursion and recurrences**	Recursive definitions, linear recurrences, characteristic equations	Solving recurrences, closed forms
8	**Discrete probability**	Sample spaces, conditional probability, expectation (brief)	Simple probability and expectation calculations
9	**Graph theory I**	Graphs, paths, connectivity, trees	Degrees, connected components, spanning trees
10	**Graph theory II**	Euler and Hamilton paths, coloring, matching	Chromatic numbers, Euler circuits

Notes. Follows the six topics of HKUST COMP2711 (enumeration, number theory, logic and proofs, recursion, probability, graph theory). The probability lecture is brief; Probability I goes much deeper.

Proof. Proofs will be added later.

QNT-103 · Math

Linear Algebra I

Stage 1
10 lectures x 3 h

Vectors, matrices, systems of equations, determinants and eigenvalues.

Prerequisites	None
University courses	HKU MATH2101 · CUHK MATH1030 · HKUST MATH2121
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Solve linear systems
- Compute determinants and eigenvalues
- Work with vector spaces and linear maps

Course outline

#	Topic	Theory	Computation / Coding
1	**Systems of linear equations**	Gauss-Jordan elimination, consistency, free variables	Solving systems, parametric solutions
2	**Matrices**	Matrix operations, inverses, elementary matrices	Products, inverses, row reduction
3	**Vector geometry**	Vectors in R^n, dot and cross products, lines and planes	Angles, projections, distance to a plane
4	**Determinants**	Cofactor expansion, properties, determinant and invertibility	Determinant values, area and volume
5	**Vector spaces and subspaces**	Axioms, subspaces, examples beyond R^n	Testing subspaces
6	**Spanning and linear independence**	Span, dependence and independence	Testing independence, spanning sets
7	**Basis and dimension**	Basis, dimension, coordinates, change of basis	Coordinates relative to a basis
8	**Linear transformations**	Kernel, image, matrix representation	Matrix of a transformation, kernel and image
9	**Row space, column space and rank**	Row, column and null spaces, rank-nullity theorem	Rank, nullity, bases of the fundamental subspaces
10	**Eigenvalues and eigenvectors**	Characteristic polynomial, eigenspaces, diagonalization basics	Eigenvalues, eigenvectors, powers of a matrix

Notes. Follows HKU MATH2101 (linear equations and matrices, vector geometry, vector spaces and linear transformations) and the CUHK MATH1030 schedule, which also reaches determinants and eigenvalues. Eigenvalues are introduced here and developed in Linear Algebra II.

Proof. Proofs will be added later.

QNT-104 · Math

Differential Equations

Stage 2
11 lectures x 3 h

Ordinary differential equations: solution methods, linear systems and stability.

Prerequisites	QNT-101 Calculus, QNT-103 Linear Algebra I
University courses	HKU MATH3405 · CityU MA3511
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Solve first and second-order ODEs
- Handle linear systems via eigenvalues
- Analyse stability and use Laplace transforms

Course outline

#	Topic	Theory	Computation / Coding
1	**First-order ODEs I**	Separable and linear equations, integrating factors	Solving separable and linear equations
2	**First-order ODEs II**	Homogeneous and exact equations, existence and uniqueness	Exact equations, initial value problems
3	**Higher-order linear ODEs**	Initial value problems, Wronskian, linear independence, reduction of order	Wronskians, reduction of order
4	**Constant coefficients**	Characteristic equations, undetermined coefficients	General and particular solutions
5	**Variation of parameters**	Variation of parameters, comparison with undetermined coefficients	Particular solutions by variation of parameters
6	**Series solutions**	Power series solutions of second-order linear equations	Series coefficients, recurrence relations
7	**Euler equations and Bessel's equation**	Regular singular points, Euler equations, Bessel's equation	Solving Euler equations
8	**Systems of ODEs**	Matrix methods, eigenvalues and eigenvectors	Solving linear systems
9	**Phase portraits and stability**	Equilibria, phase portraits, stability	Classifying equilibria
10	**Fourier series**	Fourier coefficients, convergence, even and odd extensions	Fourier coefficients of simple functions
11	**Separation of variables for PDEs**	Heat equation and wave equation by separation of variables	Solutions of the heat equation with boundary conditions

Notes. Follows the five topics of CityU MA3511: first-order ODEs, higher-order linear ODEs, series solutions, systems with phase portraits, and Fourier series with separation of variables for PDEs.

Proof. Proofs will be added later.

QNT-105 · Math

Linear Algebra II

Stage 2
10 lectures x 3 h

Inner product spaces, spectral theory, SVD and applications to PCA and data analysis.

Prerequisites	QNT-103 Linear Algebra I
University courses	CUHK MATH2040A · HKU Linear Algebra II (syllabus not verified); plus SVD / PCA applications
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Orthogonalise and project
- Apply the spectral theorem and SVD
- Connect linear algebra to PCA and least squares

Course outline

#	Topic	Theory	Computation / Coding
1	**Abstract vector spaces**	Function and polynomial spaces, bases and dimension	Coordinates in function spaces
2	**Linear transformations and change of basis**	Kernel, image, matrix representation, change of basis, similarity	Matrix of an operator in a new basis
3	**Eigenvalues and the characteristic polynomial**	Eigenvalues and eigenvectors of operators, algebraic and geometric multiplicity	Characteristic polynomials, eigenspaces
4	**Diagonalizability**	Conditions for diagonalization, matrix powers	Diagonalizing a matrix, computing A^n
5	**Inner product spaces**	Inner products, norms, Cauchy-Schwarz	Inner products, angles, norms
6	**Orthogonality and least squares**	Gram-Schmidt, orthogonal projection, least squares	Orthonormal bases, least squares solutions
7	**Adjoint, normal and self-adjoint operators**	Adjoint operator, normal and self-adjoint operators	Adjoints, checking normality
8	**Spectral theorems**	Real and complex spectral theorems, orthogonal diagonalization	Orthogonally diagonalizing a symmetric matrix
9	**Quadratic forms and definiteness**	Quadratic forms, positive definite matrices, covariance matrices	Classifying forms, checking definiteness
10	**Singular value decomposition and PCA**	SVD, low-rank approximation, principal components	SVD of a small matrix, PCA directions

Notes. Follows CUHK MATH2040A (abstract vector spaces, eigenvalues, inner product spaces, spectral theorems), without Jordan form, plus SVD and PCA for applications. The HKU Linear Algebra II syllabus was not verified.

Proof. Proofs will be added later.

QNT-106 · Math

Multivariable Calculus

Stage 2
10 lectures x 3 h

Calculus in several variables: partial derivatives, optimisation, multiple integrals and vector calculus.

Prerequisites	QNT-101 Calculus
University courses	HKU MATH2014 · CUHK MATH2010 · HKUST MATH2011 · CityU MA2001
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Compute gradients and Jacobians
- Solve constrained optimisation
- Evaluate multiple and line/surface integrals

Course outline

#	Topic	Theory	Computation / Coding
1	**Parametric and polar curves**	Parametric equations, polar coordinates, calculus in polar coordinates	Slopes, arc lengths and areas in parametric and polar form
2	**Vectors in the plane and in space**	Dot and cross products, lines and planes	Angles, projections, plane equations
3	**Vector-valued functions**	Curves in space, motion, arc length, curvature and normal vectors	Velocity, acceleration, curvature
4	**Functions of several variables**	Surfaces, graphs and level curves, limits and continuity	Level curves, limits in two variables
5	**Partial derivatives and the chain rule**	Partial derivatives, higher-order derivatives, the multivariable chain rule	Partial derivatives, chain rule evaluations
6	**Gradient, tangent planes and linear approximation**	Directional derivatives, gradient, tangent planes, linearization	Directional derivatives, tangent plane equations
7	**Maxima and minima**	Critical points, second derivative test, Lagrange multipliers	Local extrema, constrained optimization
8	**Double integrals**	Rectangular and general regions, polar coordinates	Double integrals, areas and volumes
9	**Triple integrals and change of variables**	Cylindrical and spherical coordinates, the Jacobian	Triple integrals, Jacobian determinants
10	**Vector fields and line integrals**	Vector fields, line integrals, conservative fields, Green's theorem	Line integrals, work, Green's theorem

Notes. Follows the five topics of HKUST MATH2011. Lagrange multipliers and the Jacobian change of variables are added, because optimization and probability both need them.

Proof. Proofs will be added later.

QNT-107 · Math

Numerical Methods I

Stage 2
10 lectures x 3 h

Numerical linear algebra and root finding, with Python implementations.

Prerequisites	QNT-101 Calculus, QNT-103 Linear Algebra I, QNT-705 Python
University courses	CUHK MATH3230
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Solve linear systems stably
- Find roots and fit data
- Understand error and conditioning

Course outline

#	Topic	Theory	Computation / Coding
1	**Floating-point arithmetic and error**	Roundoff, significant digits, cancellation, stability	Roundoff error and relative error calculations
2	**Root finding I**	Bisection, fixed-point iteration, convergence conditions	Iterations of bisection and fixed point **Coding:** Implement bisection and fixed-point iteration
3	**Root finding II**	Newton's method, secant method, order of convergence	Newton iterations, convergence order **Coding:** Implement Newton and secant methods
4	**Interpolation I**	Lagrange polynomials, interpolation error	Interpolating polynomials, error bounds **Coding:** Evaluate a Lagrange interpolant
5	**Interpolation II**	Divided differences, Newton form, cubic splines	Divided-difference tables **Coding:** Build a Newton interpolating polynomial
6	**Numerical differentiation**	Forward, backward and central differences, error orders	Finite difference values, error orders **Coding:** Finite difference with a step-size study
7	**Numerical integration**	Trapezoid and Simpson rules, composite rules, error terms	Composite rule approximations, error bounds **Coding:** Composite Simpson integrator
8	**Linear systems I**	Gaussian elimination, partial pivoting, LU factorization	LU factors, solving by substitution **Coding:** LU factorization with pivoting
9	**Linear systems II**	Jacobi, Gauss-Seidel and SOR iterations, condition numbers	Iterations of Jacobi and Gauss-Seidel, condition numbers **Coding:** Gauss-Seidel solver
10	**Euler's method for ODEs**	Euler's method, local and global error	Euler steps, error estimates **Coding:** Euler solver for an initial value problem

Notes. No syllabus was available for CUHK MATH3230, so this follows HKUST MATH3312: computer arithmetic, root finding, interpolation, numerical differentiation and integration, Euler's method, and linear systems. Conditioning and splines are added. Python is a prerequisite, so coding questions are included.

Proof. Proofs will be added later.

QNT-108 · Math

Real Analysis

Stage 2
10 lectures x 3 h

Rigorous foundations of calculus: sequences, continuity, differentiation and integration.

Prerequisites	QNT-101 Calculus
University courses	CityU MA4542
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Prove convergence and continuity statements
- Work with uniform convergence
- Understand Riemann and basic Lebesgue integration

Course outline

#	Topic	Theory	Computation / Coding
1	**Review of the real line**	Countability, open and closed sets, limits of sets	Countability and set-limit exercises
2	**Sigma-algebras and measurable sets**	Sigma-algebras, Borel sets, measurable sets	Generating sigma-algebras, set operations
3	**Lebesgue outer measure**	Outer measure, covering by intervals, Caratheodory criterion	Outer measures of simple sets
4	**Lebesgue measure**	Countable additivity, continuity of measure, null sets	Measures of unions and limits of sets
5	**Measurable functions**	Definition, operations, limits of measurable functions	Checking measurability
6	**The Lebesgue integral**	Simple functions, integral of nonnegative functions	Integrals of simple functions
7	**Convergence theorems**	Monotone convergence, Fatou's lemma, dominated convergence	Limits of integrals
8	**Lebesgue vs Riemann integrals**	Riemann integrability, comparison, absolute continuity	Cases where the two integrals agree or differ
9	**Lp spaces**	Holder and Minkowski inequalities, completeness of Lp	Lp norms of simple functions
10	**Integration on measure spaces**	General measure spaces, product measures, Fubini's theorem	Iterated integrals, applying Fubini

Notes. Follows the six topics of CityU MA4542: measurable sets, measurable functions, Lebesgue measure, Lebesgue integrals, Lp spaces, integration on measure spaces. This is the measure-theory base for Probability III.

Proof. Proofs will be added later.

QNT-109 · Math

Numerical Methods II

Stage 3
10 lectures x 3 h

Numerical differential equations, quadrature and PDE methods, with Python implementations.

Prerequisites	QNT-107 Numerical Methods I, QNT-104 Differential Equations
University courses	CUHK MATH3240 · CityU MA3514
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Integrate ODEs numerically
- Discretise simple PDEs
- Analyse stability and convergence

Course outline

#	Topic	Theory	Computation / Coding
1	**Initial value problems I**	Euler's method, local and global truncation error, consistency	Truncation error orders **Coding:** Euler and improved Euler solvers
2	**Initial value problems II**	Runge-Kutta methods, order conditions	Runge-Kutta steps **Coding:** Fourth-order Runge-Kutta solver
3	**Initial value problems III**	Multistep methods, stability regions, stiff equations	Stability regions, step size limits **Coding:** Adams-Bashforth solver
4	**Systems of ODEs and adaptive steps**	Systems of ODEs, adaptive step size, error control	Step size selection **Coding:** Adaptive Runge-Kutta solver
5	**Finite differences for boundary value problems**	Two-point boundary value problems, discretization	Setting up and solving the linear system **Coding:** Finite difference solver for a boundary value problem
6	**Convergence and stability of schemes**	Consistency, stability, convergence, Lax equivalence	Order of accuracy from grid refinement **Coding:** Convergence study of a scheme
7	**Finite differences for PDEs I**	Heat equation: explicit, implicit, Crank-Nicolson	Steps of the explicit and implicit schemes **Coding:** Crank-Nicolson heat equation solver
8	**Finite differences for PDEs II**	Von Neumann stability, wave and Poisson equations	Stability conditions **Coding:** Solver for the Poisson equation
9	**Finite element method I**	Weak form, Galerkin method, piecewise linear basis functions	Weak form and element matrices **Coding:** Assemble a 1D stiffness matrix
10	**Finite element method II**	Assembly, boundary conditions, error estimates	Assembled system, error estimates **Coding:** 1D finite element solver

Notes. Follows CityU MA3514 (initial value problems, finite differences for boundary value problems and PDEs, finite elements for boundary value problems). Runge-Kutta, multistep, stability analysis and adaptive step size are filled in. Coding questions are included.

Proof. Proofs will be added later.

QNT-110 · Math

Optimization

Stage 3
10 lectures x 3 h

Linear, convex and nonlinear optimisation with duality and optimality conditions.

Prerequisites	QNT-106 Multivariable Calculus, QNT-103 Linear Algebra I
University courses	CityU MS3601 · CUHK MATH3215
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Formulate and solve LPs
- Apply KKT conditions and duality
- Recognise convex problems

Course outline

#	Topic	Theory	Computation / Coding
1	**Linear programming models**	Formulating problems, feasible regions, the graphical method	Formulations, optimal vertices
2	**Simplex method I**	Basic feasible solutions, pivoting, the simplex tableau	One-pivot and full simplex iterations
3	**Simplex method II**	Degeneracy, unboundedness, two-phase method	Two-phase problems
4	**Duality**	Primal and dual problems, weak and strong duality, complementary slackness	Writing the dual, dual solutions
5	**Sensitivity analysis**	Shadow prices, ranging of coefficients and right-hand sides	Shadow prices, allowable ranges
6	**Network flows and applications**	Network flow models, foreign exchange, investment decisions	Max flow, min cost flow, arbitrage cycles
7	**Integer programming I**	Formulations, branch-and-bound	Branch-and-bound trees
8	**Integer programming II**	Cutting planes, complexity	Gomory cuts, bound comparisons
9	**Convex optimization and optimality conditions**	Convex sets and functions, Lagrangian duality, KKT conditions	Verifying convexity, solving KKT systems
10	**Quadratic programming and portfolio optimization**	Quadratic programs, mean-variance portfolios, clustering	Minimum-variance portfolios

Notes. Follows the four parts of CityU MS3601: linear programming, integer programming, convex optimization, and applications such as network flows and portfolio optimization. Computation is by hand on small problems, since Python is not a prerequisite here.

Proof. Proofs will be added later.

QNT-111 · Math

Information Theory

Stage 4
11 lectures x 3 h

Entropy, mutual information, source and channel coding.

Prerequisites	QNT-202 Probability II
University courses	CUHK Information Theory (Institute of Network Coding; course code not found)
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Compute entropy and mutual information
- Apply source and channel coding theorems
- Connect information theory to statistics

Course outline

#	Topic	Theory	Computation / Coding
1	**The science of information**	Uncertainty, bits, why entropy measures information	Entropy of simple sources
2	**Information measures**	Entropy, joint and conditional entropy, mutual information, KL divergence	Computing the information measures of a joint distribution
3	**The I-measure**	Information diagrams, set-theoretic view of information measures	Reading identities from information diagrams
4	**Zero-error data compression**	Kraft inequality, Huffman codes, optimal code lengths	Building a Huffman code, expected length
5	**Weak typicality**	Asymptotic equipartition property, source coding theorem	Typical set sizes and probabilities
6	**Strong typicality**	Strongly typical sequences, joint typicality	Counting typical sequences
7	**Discrete memoryless channels**	Channel capacity, channel coding theorem	Capacity of the binary symmetric and erasure channels
8	**Rate-distortion theory**	Distortion measures, the rate-distortion function	Rate-distortion function of simple sources
9	**The Blahut-Arimoto algorithms**	Alternating optimization for capacity and rate-distortion	Hand iterations of Blahut-Arimoto
10	**Differential entropy**	Differential entropy, maximum entropy distributions	Differential entropy of uniform, exponential and normal variables
11	**Continuous-valued channels**	Gaussian channel capacity, parallel channels, water-filling	Capacity of a Gaussian channel, water-filling allocation

Notes. Follows the 11 topics of the CUHK information theory course based on Yeung's Information Theory and Network Coding (prerequisite: undergraduate probability).

Proof. Proofs will be added later.

QNT-201 · Probability & Statistics

Probability I

Stage 2
12 lectures x 3 h

The foundations of probability: events, conditioning, discrete random variables and the classical distributions.

Prerequisites	QNT-101 Calculus
University courses	HKU SDST2601 · HKUST MATH2421 · CityU MA2510
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Count outcomes and compute probabilities of events
- Use conditioning and Bayes' theorem
- Work with expectation, variance and named discrete distributions

Course outline

#	Topic	Theory	Computation / Coding
1	**Sample spaces and event operations**	Set operations, De Morgan's laws, mutually exclusive and exhaustive events	Simplifying event expressions, counting with permutations and combinations
2	**Probability and its laws**	Axioms, addition rule, inclusion-exclusion	Probabilities with equally likely outcomes
3	**Conditional probability**	Multiplication rule, law of total probability, Bayes' theorem	Tree diagrams, test accuracy problems
4	**Independence**	Pairwise vs mutual independence, conditional independence	Combinations of independent events, system reliability
5	**Discrete random variables**	pmf, cdf, expectation, variance, linearity	Computing the cdf, mean and variance from a pmf
6	**Common discrete distributions**	Bernoulli, binomial, geometric, Poisson (limit of the binomial)	Distribution probabilities, Poisson approximation
7	**Continuous random variables**	pdf, cdf, expectation and variance	Probabilities by integration, quantiles
8	**Exponential, gamma, normal**	Memorylessness, standardisation, the gamma-exponential relationship	Normal table lookups, waiting times
9	**Functions of a random variable**	cdf method, change of variables	Finding the pdf or pmf of a transformed variable
10	**Joint, marginal, conditional distributions**	Discrete and continuous cases, independent random variables	Marginal and conditional distributions from a joint pdf
11	**Functions of jointly distributed variables**	Sums, maximum and minimum, convolution	Distribution of a sum of two variables
12	**Expectation, variance, covariance, correlation**	Variance of a sum, what correlation means	Variance of a two-asset portfolio

Notes. Follows the 22 topics of HKU SDST2601. Expectation and variance of a single variable move up to lecture 5, because the named distributions need them.

Proof. Proofs will be added later.

QNT-202 · Probability & Statistics

Probability II

Stage 3
11 lectures x 3 h

Continuous and multivariate random variables, transformations, conditional expectation, and the limit theorems that underlie all of statistics.

Prerequisites	QNT-201 Probability I, QNT-106 Multivariable Calculus
University courses	HKU SDST2602 · CUHK STAT2006
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Work with joint, marginal and conditional densities
- Transform random variables and derive their distributions
- Use MGFs, conditioning and the CLT to solve problems

Course outline

#	Topic	Theory	Computation / Coding
1	**Continuous random variables**	pdf, cdf, expectation, variance of continuous variables	Integrating densities, computing moments
2	**Named continuous distributions**	Uniform, exponential, gamma, beta, normal; memoryless property	Normal table lookups, exponential waiting times
3	**Joint distributions**	Joint pmf/pdf, marginals, independence	Double integrals over regions, marginal densities
4	**Conditional distributions**	Conditional pdf, conditional expectation and variance	E[X **Coding:** Y] calculations, law of total expectation and variance
5	**Covariance and correlation**	Covariance, correlation, variance of sums	Covariance matrices of small vectors
6	**Transformations of variables**	CDF method, Jacobian method, convolution	Densities of Y=g(X), sums of independent variables
7	**Order statistics**	Distribution of min, max and k-th order statistic	Min/max of exponential and uniform samples
8	**Moment generating functions**	MGF, uniqueness, sums of independent variables	Identifying distributions from MGFs
9	**Multivariate normal**	Definition, linear transformations, conditional distribution	Conditional mean and variance of bivariate normals
10	**Modes of convergence**	In probability, in distribution, almost surely (intuition)	Checking convergence of simple sequences
11	**Law of large numbers and CLT**	Markov, Chebyshev, WLLN, CLT	Normal approximations to binomial and sums

Notes. Follows HKU SDST2602 (multivariate and continuous probability). Assumes the discrete named distributions from Probability I.

Proof. Proofs will be added later.

QNT-203 · Probability & Statistics

Regression I

Stage 3
11 lectures x 3 h

The linear model fitted by least squares, with inference, diagnostics and model selection.

Prerequisites	QNT-201 Probability I, QNT-103 Linear Algebra I
University courses	HKU ECON2280 · CUHK STAT3008 · CityU MS3252
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Fit and interpret linear models
- Run inference on coefficients
- Diagnose and repair model violations

Course outline

#	Topic	Theory	Computation / Coding
1	**Simple linear regression**	Model, least squares, interpretation	Computing slope and intercept from data
2	**Properties of OLS**	Gauss-Markov, unbiasedness, variance of estimators	Standard errors from summary statistics
3	**Inference in simple regression**	t tests, intervals, prediction intervals	Testing slopes, building intervals
4	**Matrix formulation**	Normal equations, projection (hat) matrix	Solving small normal equations by hand
5	**Multiple regression**	Partial effects, R-squared, adjusted R-squared	Interpreting coefficient tables
6	**F tests and nested models**	Extra sum of squares, general linear hypotheses	ANOVA table calculations
7	**Dummy variables and interactions**	Categorical predictors, interaction terms	Reading coefficients with dummies
8	**Residual diagnostics**	Residual plots, leverage, Cook's distance	Computing leverage and influence
9	**Heteroskedasticity and correlated errors**	Robust SEs, WLS, GLS	Weighted least squares examples
10	**Multicollinearity and variable selection**	VIF, AIC, BIC, stepwise methods	Computing VIF, comparing models by AIC
11	**Transformations and prediction**	Box-Cox, log models, cross-validation	Back-transformed predictions

Notes. Follows HKU ECON2280 / CUHK STAT3008 content on linear regression.

Proof. Proofs will be added later.

QNT-204 · Probability & Statistics

Monte Carlo

Stage 4
11 lectures x 3 h

Simulation-based computation: random number generation, variance reduction, MCMC and bootstrap.

Prerequisites	QNT-202 Probability II, QNT-705 Python
University courses	CUHK RMSC4001 · HKU STAT8309
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Generate samples from arbitrary distributions
- Reduce variance of Monte Carlo estimators
- Implement MCMC and bootstrap

Course outline

#	Topic	Theory	Computation / Coding
1	**Random number generation**	LCGs, quality of PRNGs, seeding	Check a generator's statistics **Coding:** Implement an LCG and a seeded sampler
2	**Inverse transform and acceptance-rejection**	Inverse CDF, rejection envelope	Efficiency of rejection samplers **Coding:** Sample from a custom density
3	**Special methods**	Box-Muller, composition, multivariate normal	Cholesky-based sampling **Coding:** Generate correlated normals
4	**Monte Carlo integration**	Estimator, standard error, CLT	Confidence intervals for MC estimates **Coding:** Estimate an integral with error bars
5	**Variance reduction I**	Antithetic variates, control variates	Variance reduction factors **Coding:** Implement antithetic and control variates
6	**Variance reduction II**	Importance sampling, stratification	Optimal importance densities **Coding:** Rare-event estimation
7	**Discretising SDEs**	Euler and Milstein schemes, weak/strong error	Convergence orders **Coding:** Simulate GBM paths
8	**Markov chain Monte Carlo**	Metropolis-Hastings, detailed balance	Acceptance ratios **Coding:** Implement random-walk Metropolis
9	**Gibbs sampling**	Full conditionals, blocking	Deriving conditionals **Coding:** Gibbs sampler for a bivariate normal
10	**MCMC diagnostics**	Burn-in, ESS, R-hat	Computing ESS **Coding:** Diagnose chains from output
11	**Bootstrap and resampling**	Nonparametric bootstrap, percentile intervals	Bootstrap intervals by hand **Coding:** Bootstrap CI for a statistic

Notes. Follows CUHK RMSC4001 style, with coding emphasised.

Proof. Proofs will be added later.

QNT-205 · Probability & Statistics

Probability III

Stage 4
11 lectures x 3 h

Probability rebuilt on measure theory: sigma-algebras, expectation as an integral, convergence theorems, conditional expectation and martingales.

Prerequisites	QNT-202 Probability II, QNT-108 Real Analysis
University courses	HKU MATH3603 · HKUST MAFS5020
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- State probability in measure-theoretic language
- Apply the convergence theorems and Borel-Cantelli
- Use conditional expectation and martingales

Course outline

#	Topic	Theory	Computation / Coding
1	**Sigma-algebras and probability measures**	Measurable spaces, probability spaces, generated sigma-algebras	Verifying a collection is a sigma-algebra
2	**Random variables as measurable functions**	Measurability, distribution of a random variable	Pushforward measures of simple maps
3	**Expectation as a Lebesgue integral**	Simple functions, monotone convergence, Fatou	Computing expectations via the definition
4	**Convergence theorems**	Dominated convergence, uniform integrability	Limit-interchange exercises
5	**Independence and Borel-Cantelli**	Independent sigma-algebras, both Borel-Cantelli lemmas	Probabilities of limsup events
6	**Modes of convergence**	a.s., in probability, L^p, in distribution and their relations	Classifying convergence of examples
7	**Laws of large numbers**	Weak and strong LLN	Applications and counter-examples
8	**Characteristic functions**	Definition, inversion, Levy continuity	Characteristic functions of standard laws
9	**Central limit theorem**	Lindeberg-Feller CLT	Checking the Lindeberg condition
10	**Conditional expectation**	Definition via Radon-Nikodym, tower property	Conditional expectations given sigma-algebras
11	**Martingales**	Martingales, stopping times, optional stopping	Martingale checks, gambler's ruin

Notes. Measure-theoretic probability, following the HKU MATH3603 / HKUST MAFS5020 style. Requires Real Analysis.

Proof. Proofs are the core of this topic, so Proof carries more weight here.

QNT-206 · Probability & Statistics

Statistical Inference

Stage 4
11 lectures x 3 h

Estimation, testing and confidence intervals from a unified likelihood-based viewpoint.

Prerequisites	QNT-202 Probability II
University courses	HKU STAT3602 · HKUST MATH3423 · CityU MS2602
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Derive and compare estimators
- Construct tests and intervals
- Judge optimality via sufficiency and the Cramer-Rao bound

Course outline

#	Topic	Theory	Computation / Coding
1	**Samples and sampling distributions**	Statistics, chi-square, t and F distributions	Distribution of sample mean and variance
2	**Method of moments**	Moment equations, consistency	Moment estimators for common families
3	**Maximum likelihood**	Likelihood, score, invariance	Closed-form and numeric MLEs
4	**Properties of estimators**	Bias, MSE, consistency, efficiency	Comparing estimators by MSE
5	**Sufficiency and completeness**	Factorisation theorem, Rao-Blackwell, Lehmann-Scheffe	Finding sufficient statistics and UMVUEs
6	**Fisher information and Cramer-Rao**	Information, CR bound, asymptotic normality of MLE	Computing information and bounds
7	**Confidence intervals**	Pivots, intervals for mean, variance, proportions	Interval construction and interpretation
8	**Hypothesis testing basics**	Type I/II errors, power, p-values	Power calculations
9	**Neyman-Pearson and likelihood ratio tests**	NP lemma, LRT, Wilks theorem	Constructing LRTs, critical regions
10	**Tests for means and variances**	One- and two-sample t, F tests	Test statistics and decisions from summaries
11	**Chi-square tests**	Goodness of fit, independence	Contingency table tests

Notes. Follows HKU STAT3602. Assumes Probability II including MGFs and the CLT.

Proof. Proofs will be added later.

QNT-207 · Probability & Statistics

Stochastic Processes

Stage 4
11 lectures x 3 h

Random processes: Markov chains, Poisson processes, queues and branching.

Prerequisites	QNT-202 Probability II
University courses	HKU SDST3603 · CUHK STAT3007 · HKUST ISOM3540
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Classify Markov chains and compute stationary distributions
- Model arrivals with Poisson processes
- Analyse simple queues and martingale problems

Course outline

#	Topic	Theory	Computation / Coding
1	**Markov chains: basics**	Transition matrices, Chapman-Kolmogorov	n-step probabilities
2	**Classification of states**	Communication, recurrence, periodicity	Classifying states from a diagram
3	**Hitting probabilities and times**	First-step analysis	Solving absorption problems
4	**Stationary distributions**	Existence, uniqueness, limit theorems	Solving pi P = pi
5	**Random walks and gambler's ruin**	Ruin probabilities, expected duration	Ruin probability calculations
6	**Poisson process**	Interarrival times, superposition, thinning	Event probabilities in time windows
7	**Continuous-time Markov chains**	Generators, Kolmogorov equations	Solving small generator systems
8	**Birth-death processes and queues**	M/M/1, M/M/c, Little's law	Queue length and waiting time formulas
9	**Renewal theory**	Renewal function, renewal reward theorem	Long-run average rewards
10	**Branching processes**	Extinction probability, generating functions	Computing extinction probabilities
11	**Martingales (discrete)**	Definition, optional stopping	Applying optional stopping

Notes. Follows HKU SDST3603 / CUHK STAT3007. Discrete-time chains first, then continuous time.

Proof. Proofs will be added later.

QNT-208 · Probability & Statistics

Bayesian

Stage 5
12 lectures x 3 h

Bayesian inference from conjugate models to MCMC, with hierarchical modelling and model checking.

Prerequisites	QNT-206 Statistical Inference, QNT-705 Python
University courses	HKU STAT4610 · CUHK STAT4010
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Derive posteriors for standard models
- Implement MCMC samplers
- Compare models and check fit

Course outline

#	Topic	Theory	Computation / Coding
1	**Bayesian framework**	Likelihood, prior, posterior, posterior predictive	Posterior for a discrete parameter
2	**Conjugate models**	Beta-binomial, normal-normal, gamma-Poisson	Posterior updates, shrinkage weights
3	**Posterior summaries**	MAP, posterior mean, credible vs HPD vs confidence intervals	A 95% credible interval
4	**Choosing a prior**	Elicitation, noninformative and weakly informative priors, sensitivity analysis	Comparing posteriors under different priors
5	**Model evaluation and selection**	Posterior predictive checks, Bayes factors, WAIC	Comparing two models
6	**Monte Carlo basics**	Law of large numbers, bias and variance, inverse transform sampling	Standard error, required sample size **Coding:** Inverse transform sampler
7	**Importance sampling**	Proposal choice, weights, effective sample size	Computing weights and estimates **Coding:** Implement importance sampling
8	**MCMC I**	Stationary distribution, detailed balance, Metropolis-Hastings	Acceptance rates **Coding:** Metropolis-Hastings sampler
9	**MCMC II**	Gibbs sampling, hierarchical models, burn-in, ESS, R-hat, HMC overview	Diagnostic statistics **Coding:** Gibbs sampler
10	**Laplace approximation**	Normal approximation of the posterior, when it applies	Computing a Laplace approximation
11	**Variational Bayes**	ELBO, mean-field, comparison with MCMC	The ELBO of a simple model
12	**Nonparametric Bayes**	Dirichlet process, Gaussian process, machine learning applications	A simple Dirichlet process calculation

Notes. Follows HKU SDST4610 (Bayesian learning). Coding questions are graded with a fixed random seed and a statistical tolerance.

Proof. Proofs will be added later.

QNT-209 · Probability & Statistics

Causal Inference

Stage 5
10 lectures x 3 h

Identifying causal effects from observational and experimental data.

Prerequisites	QNT-203 Regression I, QNT-206 Statistical Inference
University courses	HKU SDST4613 · HKU ECON3284 · HKUST IEDA4000D
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Formulate causal questions with potential outcomes
- Pick an identification strategy
- Estimate effects and check assumptions

Course outline

#	Topic	Theory	Computation / Coding
1	**Potential outcomes**	Rubin model, ATE, SUTVA	Computing ATE from a table
2	**Randomised experiments**	Randomisation inference, Neyman estimator	Difference-in-means with SE
3	**Confounding and DAGs**	Causal graphs, backdoor criterion	Choosing adjustment sets
4	**Adjustment and matching**	Regression adjustment, matching	Matched estimates by hand
5	**Propensity scores**	Score theorem, IPW, doubly robust	IPW estimator calculations
6	**Instrumental variables**	Relevance, exclusion, LATE, 2SLS	Wald estimator, 2SLS by hand
7	**Regression discontinuity**	Sharp and fuzzy RD	Local linear estimates at a cutoff
8	**Difference-in-differences**	Parallel trends, event studies	2x2 DiD estimates
9	**Synthetic control and panel methods**	Synthetic control, fixed effects	Weights and effect estimates
10	**Sensitivity and heterogeneity**	Rosenbaum bounds, CATE intro	Bounding effects under confounding

Notes. Follows HKU SDST4613 and ECON3284 style (potential outcomes and econometric designs).

Proof. Proofs will be added later.

QNT-210 · Probability & Statistics

Experimental Design / A/B Testing

Stage 5
11 lectures x 3 h

Designing experiments and running A/B tests that give trustworthy answers.

Prerequisites	QNT-206 Statistical Inference, QNT-203 Regression I
University courses	CityU DSC4110 (mostly industrial experimental design; no university course covers the A/B testing part)
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Design randomised experiments
- Size tests and analyse results
- Avoid peeking and multiple-testing traps

Course outline

#	Topic	Theory	Computation / Coding
1	**Principles of experimentation**	Randomisation, replication, blocking	Identifying flaws in designs
2	**One-way ANOVA**	Model, F test, multiple comparisons	ANOVA table calculations
3	**Factorial designs**	Main effects, interactions, 2^k designs	Effects from 2^2 and 2^3 tables
4	**Blocking and Latin squares**	Randomised blocks, Latin squares	Block-adjusted comparisons
5	**Fractional factorials**	Aliasing, resolution	Alias structures
6	**A/B test basics**	Hypotheses, metrics, randomisation units	Choosing metrics and units
7	**Sample size and power**	MDE, power analysis	Sample size calculations
8	**Inference for proportions and means**	z/t tests, delta method for ratios	Test statistics for A/B summaries
9	**Sequential testing and peeking**	Alpha spending, always-valid inference	Adjusted boundaries
10	**Variance reduction**	CUPED, stratification	Variance reduction from covariates
11	**Pitfalls and interference**	SRM, novelty effects, network interference	Diagnosing sample ratio mismatch

Notes. Based on CityU DSC4110 for classical design and standard industry A/B practice; no single university course covers online experiments.

Proof. Proofs will be added later.

QNT-211 · Probability & Statistics

Nonparametric

Stage 5
10 lectures x 3 h

Inference without strong distributional assumptions: rank tests, permutation methods and smoothing.

Prerequisites	QNT-206 Statistical Inference
University courses	HKU STAT3620
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Choose and run rank-based tests
- Use permutation and bootstrap inference
- Estimate densities and regression curves nonparametrically

Course outline

#	Topic	Theory	Computation / Coding
1	**Why nonparametric**	Parametric vs distribution-free, efficiency	Comparing test power
2	**Sign and signed-rank tests**	Sign test, Wilcoxon signed-rank	Exact and normal-approximation tests
3	**Two-sample rank tests**	Mann-Whitney, Wilcoxon rank-sum	Computing U and p-values
4	**Multi-sample tests**	Kruskal-Wallis, Friedman	H statistics from rank sums
5	**Tests of fit**	Kolmogorov-Smirnov, Anderson-Darling	KS statistic from small samples
6	**Rank correlation**	Spearman, Kendall tau	Computing rank correlations
7	**Permutation tests**	Exchangeability, exact tests	Enumerating permutation distributions
8	**Bootstrap and jackknife**	Bootstrap variance, bias, intervals	Jackknife estimates
9	**Kernel density estimation**	Bias-variance, bandwidth selection	Density estimates at points
10	**Nonparametric regression**	Nadaraya-Watson, local linear, LOESS	Kernel regression at a point

Notes. Follows HKU STAT3620 (nonparametric methods). Distribution-free tests first, then smoothing.

Proof. Proofs will be added later.

QNT-212 · Probability & Statistics

Regression II

Stage 5
11 lectures x 3 h

Extensions of the linear model: GLMs, regularisation, mixed models and robust methods.

Prerequisites	QNT-203 Regression I, QNT-206 Statistical Inference, QNT-105 Linear Algebra II
University courses	HKUST MAFS5140
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Fit logistic, Poisson and other GLMs
- Apply ridge, lasso and elastic net
- Choose between competing regression frameworks

Course outline

#	Topic	Theory	Computation / Coding
1	**Exponential family and GLM framework**	Exponential family, link functions	Identifying canonical links
2	**Logistic regression**	Odds, MLE via IRLS, interpretation	Odds ratios and predicted probabilities
3	**Inference for GLMs**	Wald, score and deviance tests	Deviance comparisons
4	**Poisson and count models**	Poisson, overdispersion, negative binomial	Rate ratios and offsets
5	**Multinomial and ordinal models**	Softmax models, proportional odds	Class probabilities from fitted models
6	**Ridge regression**	Bias-variance, shrinkage, SVD view	Ridge solutions on small examples
7	**Lasso and sparsity**	L1 geometry, soft-thresholding, coordinate descent	Soft-thresholding calculations
8	**Elastic net and tuning**	Combined penalties, cross-validation	Choosing lambda from CV tables
9	**Splines and nonparametric regression**	Basis expansions, smoothing splines	Spline basis constructions
10	**Mixed-effects models**	Random effects, BLUPs	Variance components from small designs
11	**Robust regression**	M-estimators, Huber loss, breakdown point	Huber estimates on tiny data

Notes. Follows HKUST MAFS5140 style: generalised and regularised regression.

Proof. Proofs will be added later.

QNT-213 · Probability & Statistics

Research Methods

Stage 5
11 lectures x 3 h

How to turn an idea into a defensible quantitative result: hypotheses, backtests, validation and reproducibility.

Prerequisites	QNT-203 Regression I, QNT-301 Machine Learning
University courses	No standalone course; partly covered in HKU STAT8020 · HKUST MAFS5310
Assessment	Theory + Computation (Sufficient); Proof N/A, Complete = hardest set
Origin	Original to this app (no direct university course)

Learning outcomes

- Frame testable hypotheses
- Validate a model out of sample without leakage
- Report results honestly and reproducibly

Course outline

#	Topic	Theory	Computation / Coding
1	**From idea to hypothesis**	Hypotheses, effect sizes, null results	Writing testable hypotheses
2	**Data and leakage**	Look-ahead bias, survivorship bias	Spotting leakage in setups
3	**Train/validation/test design**	Time-series splits, walk-forward	Designing splits for given data
4	**Overfitting and multiple testing**	Data snooping, deflated Sharpe	Adjusting Sharpe for trials
5	**Backtesting**	Costs, slippage, capacity	PnL net of costs
6	**Performance metrics**	Sharpe, drawdown, information ratio	Computing metrics from returns
7	**Statistical significance in practice**	Bootstrap for Sharpe, effect vs significance	Intervals for performance
8	**Model comparison**	Nested tests, information criteria, CV	Choosing between models
9	**Robustness and sensitivity**	Parameter stability, regime tests	Sensitivity tables
10	**Reproducibility**	Version control, seeds, pipelines	Checking a reproducibility checklist
11	**Communicating results**	Paper structure, plots, limitations	Critiquing a results section

Notes. No single university course; structure follows quant research practice and partly HKU STAT8020 / HKUST MAFS5310.

Proof. Proofs are replaced by written research critiques; Proof = N/A may be considered.

QNT-214 · Probability & Statistics

Time Series I

Stage 5
10 lectures x 3 h

Classical time-series modelling with ARMA/ARIMA models, estimation and forecasting.

Prerequisites	QNT-203 Regression I, QNT-207 Stochastic Processes
University courses	HKU SDST4601 · CityU EF4822
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Identify stationary and non-stationary series
- Fit ARIMA models
- Produce and evaluate forecasts

Course outline

#	Topic	Theory	Computation / Coding
1	**Time series and stationarity**	Weak stationarity, autocovariance	Sample ACF by hand
2	**White noise and linear processes**	MA(infinity), Wold decomposition	Autocovariances of linear filters
3	**MA models**	MA(q), invertibility	ACF of MA(1), MA(2)
4	**AR models**	AR(p), causality, Yule-Walker	Solving Yule-Walker equations
5	**ARMA models**	Causal/invertible ARMA, ACF/PACF	Identifying orders from ACF/PACF
6	**Forecasting**	Best linear prediction, forecast errors	h-step forecasts and variances
7	**Estimation**	Method of moments, conditional MLE	Estimating AR(1) parameters
8	**Non-stationarity and differencing**	Unit roots, ARIMA, ADF test	Differencing and unit-root decisions
9	**Seasonal models**	SARIMA, seasonal differencing	Reading seasonal ACFs
10	**Model diagnostics**	Residual checks, Ljung-Box, AIC/BIC	Test statistics from residual ACF

Notes. Follows HKU SDST4601 / CityU EF4822 style (ARIMA family).

Proof. Proofs will be added later.

QNT-215 · Probability & Statistics

High-dimensional Stats

Stage 6
10 lectures x 3 h

Statistics when the number of variables rivals or exceeds the sample size: concentration, sparsity and random matrices.

Prerequisites	QNT-212 Regression II, QNT-105 Linear Algebra II
University courses	CUHK STAT5103
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Use concentration inequalities
- Analyse lasso and sparse recovery
- Apply random-matrix intuition to covariance estimation

Course outline

#	Topic	Theory	Computation / Coding
1	**The high-dimensional regime**	Curse of dimensionality, p>n	Distance concentration examples
2	**Sub-Gaussian variables**	Sub-Gaussian and sub-exponential tails	Tail bounds for sums
3	**Concentration inequalities**	Hoeffding, Bernstein, McDiarmid	Applying bounds to examples
4	**Covariance estimation**	Sample covariance, operator-norm error, shrinkage	Ledoit-Wolf style shrinkage
5	**Random matrices**	Marchenko-Pastur, Wigner, eigenvalue spread	MP edges for given p/n
6	**PCA in high dimensions**	Spiked models, BBP transition	Detecting spikes
7	**Sparse regression**	Lasso oracle inequalities, restricted eigenvalue	Support recovery conditions
8	**Sparse recovery**	Basis pursuit, compressed sensing, RIP	Recovery guarantees
9	**Multiple testing**	FWER, Bonferroni, FDR, Benjamini-Hochberg	BH procedure on p-values
10	**Graphical models**	Graphical lasso, precision matrices	Partial correlation reading

Notes. Follows CUHK STAT5103 style (high-dimensional statistics). Requires heavy linear algebra.

Proof. Proofs are central here, so Proof carries more weight.

QNT-216 · Probability & Statistics

Time Series II

Stage 6
10 lectures x 3 h

Heteroskedastic, multivariate and state-space models for financial time series.

Prerequisites	QNT-214 Time Series I, QNT-206 Statistical Inference
University courses	HKUST MAFS5130 · CUHK STAT6204
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Fit GARCH-type models
- Model several series jointly
- Use Kalman filtering

Course outline

#	Topic	Theory	Computation / Coding
1	**Stylised facts of returns**	Heavy tails, volatility clustering	Computing sample moments of returns
2	**ARCH models**	ARCH(q), conditional variance, estimation	Conditional variance recursion
3	**GARCH models**	GARCH(1,1), persistence, forecasting	Variance forecasts, long-run variance
4	**GARCH extensions**	EGARCH, GJR, t-innovations	Leverage-effect comparisons
5	**Realised volatility**	High-frequency estimators, microstructure noise	Realised variance from intraday data
6	**VAR models**	VAR(p), stationarity, Granger causality	Companion matrices, impulse responses
7	**Cointegration**	Engle-Granger, Johansen, VECM	Constructing spread series, error correction
8	**State-space models**	State-space form, Kalman filter	One-step Kalman updates
9	**Regime switching**	Markov switching, HMM filtering	Filtering probabilities
10	**Spectral analysis**	Periodogram, spectral density	Periodogram of small series

Notes. Follows HKUST MAFS5130 / CUHK STAT6204 style (volatility and multivariate series).

Proof. Proofs will be added later.

QNT-301 · AI / ML

Machine Learning

Stage 4
11 lectures x 3 h

Core supervised and unsupervised learning, with emphasis on generalisation and honest evaluation.

Prerequisites	QNT-203 Regression I, QNT-110 Optimization, QNT-705 Python
University courses	CUHK CSCI3320 · HKUST COMP4211 · HKUST MAFS5340 · HKU COMP7409 · CUHK RMSC4002
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Train and evaluate standard models
- Explain bias-variance and regularisation
- Build sound validation pipelines

Course outline

#	Topic	Theory	Computation / Coding
1	**Learning problems and generalisation**	Supervised/unsupervised, empirical risk, bias-variance	Train/test error computations **Coding:** Load data and fit a baseline
2	**Linear and logistic models**	Least squares, logistic loss, gradient descent	Gradient steps by hand **Coding:** Implement gradient descent
3	**Regularisation and model selection**	Ridge, lasso, cross-validation	Choosing lambda from CV output **Coding:** Implement k-fold CV
4	**k-NN and kernel methods**	Nearest neighbours, kernels, curse of dimensionality	Distance-based predictions **Coding:** Implement k-NN
5	**Support vector machines**	Margin, hinge loss, dual, kernel trick	Margins and support vectors from small data **Coding:** Fit an SVM
6	**Decision trees**	Impurity, splitting, pruning	Computing Gini and entropy splits **Coding:** Implement a tree split
7	**Ensembles**	Bagging, random forests, boosting	Bagging variance reduction **Coding:** Gradient boosting by hand
8	**Generative models**	Naive Bayes, LDA, Gaussian mixtures	Posterior class probabilities **Coding:** EM for a GMM
9	**Dimensionality reduction**	PCA, SVD, t-SNE intuition	PCA from covariance matrices **Coding:** Implement PCA
10	**Clustering**	k-means, hierarchical, spectral	k-means iterations **Coding:** Implement k-means
11	**Evaluation and finance pitfalls**	ROC/AUC, calibration, purged CV	Computing AUC and precision-recall **Coding:** Leakage-free splits

Notes. Follows CUHK CSCI3320 and HKUST COMP4211 style, with a finance-flavoured final lecture.

Proof. Proofs will be added later.

QNT-302 · AI / ML

Deep Learning

Stage 5
11 lectures x 3 h

Neural networks from backpropagation to modern architectures and training practice.

Prerequisites	QNT-301 Machine Learning
University courses	HKU COMP3340 · CUHK STAT4012 · HKU DASC7606
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Derive backprop
- Train CNNs, RNNs and Transformers
- Diagnose training problems

Course outline

#	Topic	Theory	Computation / Coding
1	**Perceptrons and MLPs**	Neurons, activations, universal approximation	Forward passes by hand **Coding:** Implement an MLP forward pass
2	**Backpropagation**	Chain rule, computational graphs	Gradients through small networks **Coding:** Implement backprop
3	**Optimisation**	SGD, momentum, Adam, learning-rate schedules	Update-rule calculations **Coding:** Implement optimisers
4	**Regularisation**	Weight decay, dropout, early stopping	Effective weight shrinkage **Coding:** Add dropout and weight decay
5	**Initialisation and normalisation**	Xavier/He init, batch/layer norm	Variance propagation **Coding:** Implement layer norm
6	**Convolutional networks**	Convolutions, pooling, receptive fields	Output size and parameter counts **Coding:** Implement convolution
7	**Modern CNNs**	ResNets, skip connections	Parameter and FLOP counts **Coding:** Build a residual block
8	**Recurrent networks**	RNNs, LSTM, GRU, vanishing gradients	Unrolled gradients **Coding:** Implement an RNN cell
9	**Attention and Transformers**	Self-attention, positional encoding	Attention weights by hand **Coding:** Implement scaled dot-product attention
10	**Generative models**	Autoencoders, VAEs, GANs	ELBO terms **Coding:** Train a small autoencoder
11	**Training practice**	Debugging, hyperparameter search, mixed precision	Diagnosing loss curves **Coding:** Overfit-one-batch test

Notes. Follows HKU COMP3340 / CUHK STAT4012 style, with PyTorch-style coding.

Proof. Proofs will be added later.

QNT-303 · AI / ML

Learning Theory

Stage 5
10 lectures x 3 h

Why learning works: PAC learning, complexity measures and generalisation bounds.

Prerequisites	QNT-301 Machine Learning, QNT-205 Probability III
University courses	CUHK CSCI4230
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- State and use PAC bounds
- Compute VC dimension and Rademacher complexity
- Connect theory to practical algorithms

Course outline

#	Topic	Theory	Computation / Coding
1	**PAC learning**	Realisable and agnostic PAC	Sample complexity for finite classes
2	**Uniform convergence**	Hoeffding and union bound	Generalisation gaps
3	**VC dimension**	Shattering, Sauer lemma	VC dimension of simple classes
4	**Rademacher complexity**	Definition, bounds	Complexity of linear classes
5	**Margin theory**	Margin bounds, SVMs	Margin-based bounds
6	**Boosting**	AdaBoost, margin explanation	Training-error bounds
7	**Online learning**	Regret, Perceptron, Hedge	Regret bounds
8	**Stability and regularisation**	Algorithmic stability	Stability constants
9	**Convex learning**	Convex Lipschitz bounded problems, SGD	SGD rates
10	**Bias-variance and modern theory**	Double descent, interpolation	Interpretation exercises

Notes. Follows CUHK CSCI4230. Needs measure-theoretic probability for some proofs.

Proof. Proofs are central; Proof carries more weight.

QNT-304 · AI / ML

MLOps

Stage 5
10 lectures x 3 h

Shipping, monitoring and maintaining models in production.

Prerequisites	QNT-301 Machine Learning, QNT-710 Databases, QNT-703 Dev Tools (Linux · Git)
University courses	No university course
Assessment	Theory + Computation (Sufficient); Proof N/A, Complete = hardest set
Origin	Original to this app (no direct university course)

Learning outcomes

- Package and deploy models
- Version data, code and experiments
- Monitor drift and failures

Course outline

#	Topic	Theory	Computation / Coding
1	**ML systems overview**	Lifecycle, failure modes	Identifying pipeline issues **Coding:** Project structure
2	**Data pipelines**	Validation, schemas, feature stores	Schema checks **Coding:** Data validation
3	**Experiment tracking**	Metrics, configs, reproducibility	Comparing runs **Coding:** Tracking runs
4	**Containers**	Docker, images, environments	Dockerfile reading **Coding:** Containerise a model
5	**Model serving**	REST/gRPC, batch vs online	Latency budgets **Coding:** Serve a model
6	**CI/CD for ML**	Tests, pipelines, rollouts	Pipeline design **Coding:** Write tests
7	**Orchestration**	DAG schedulers, retries	Dependency graphs **Coding:** Pipeline DAG
8	**Monitoring and drift**	Data and concept drift, alerts	Drift statistics **Coding:** Drift detection
9	**Scaling and cost**	GPUs, batching, cost models	Cost calculations **Coding:** Batching
10	**Governance**	Reproducibility, audit, model risk	Checklist review **Coding:** Model card

Notes. No university course; structure follows industry practice for taking models to production.

Proof. Proof = N/A; Complete = pass the hardest set.

QNT-305 · AI / ML

Reinforcement Learning

Stage 5
11 lectures x 3 h

Sequential decision making from bandits and MDPs to deep reinforcement learning.

Prerequisites	QNT-301 Machine Learning, QNT-207 Stochastic Processes
University courses	HKUST MAFS5370
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Solve MDPs by dynamic programming
- Implement TD and policy-gradient methods
- Understand RL applications in trading

Course outline

#	Topic	Theory	Computation / Coding
1	**Bandits**	Exploration vs exploitation, UCB, Thompson	Regret calculations **Coding:** Bandit simulator
2	**Markov decision processes**	States, rewards, policies, Bellman equations	Value computation **Coding:** Solve a tiny MDP
3	**Dynamic programming**	Policy/value iteration	Iterations by hand **Coding:** Value iteration
4	**Monte Carlo methods**	First-visit MC, exploration	MC returns **Coding:** MC control
5	**Temporal-difference learning**	TD(0), SARSA, Q-learning	TD updates by hand **Coding:** Tabular Q-learning
6	**n-step and eligibility traces**	TD(lambda)	Return computations **Coding:** TD(lambda)
7	**Function approximation**	Linear approximation, deadly triad	Semi-gradient updates **Coding:** Linear Q-learning
8	**Deep Q-networks**	Replay buffers, target networks	Target calculations **Coding:** DQN on a small task
9	**Policy gradients**	REINFORCE, baselines, actor-critic	Gradient estimates **Coding:** REINFORCE
10	**Modern policy methods**	PPO, TRPO, SAC	Clipped objective **Coding:** PPO objective
11	**RL in finance**	Execution, market making, reward design	Reward shaping examples **Coding:** Trading environment

Notes. Follows HKUST MAFS5370 and the standard Sutton-Barto sequence.

Proof. Proofs will be added later.

QNT-306 · AI / ML

Computer Vision

Stage 6
10 lectures x 3 h

Image understanding with classical methods and deep networks.

Prerequisites	QNT-302 Deep Learning
University courses	HKUST COMP4471
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Process and transform images
- Build detection and segmentation models
- Understand geometry and generative vision

Course outline

#	Topic	Theory	Computation / Coding
1	**Image formation and filtering**	Pixels, convolution, edges	Filter outputs **Coding:** Convolution filters
2	**Features**	Corners, SIFT, descriptors	Harris response **Coding:** Feature matching
3	**Image classification**	CNN classifiers, augmentation	Accuracy metrics **Coding:** Train a classifier
4	**Modern backbones**	ResNets, ViT	Receptive fields **Coding:** Vision Transformer patching
5	**Object detection**	R-CNN family, YOLO, IoU, NMS	IoU and NMS by hand **Coding:** Implement NMS
6	**Segmentation**	FCN, U-Net, Mask R-CNN	Dice and IoU **Coding:** Dice loss
7	**Geometry**	Camera models, homography, stereo	Projection calculations **Coding:** Homography estimation
8	**Self-supervised learning**	Contrastive learning, CLIP	Contrastive loss **Coding:** InfoNCE loss
9	**Generative vision**	GANs, diffusion models	Diffusion noise schedule **Coding:** Forward diffusion
10	**Video and deployment**	Optical flow, tracking, efficiency	Flow equations **Coding:** Basic tracker

Notes. Follows HKUST COMP4471 style.

Proof. Proofs will be added later.

QNT-307 · AI / ML

NLP

Stage 6
10 lectures x 3 h

Language modelling and text processing from n-grams to pretrained Transformers.

Prerequisites	QNT-302 Deep Learning
University courses	HKU COMP3361 · HKUST COMP4221
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Build text pipelines
- Understand embeddings and sequence models
- Fine-tune and evaluate pretrained models

Course outline

#	Topic	Theory	Computation / Coding
1	**Text processing**	Tokenisation, normalisation, BPE	Applying BPE merges **Coding:** Implement BPE
2	**N-gram language models**	Perplexity, smoothing	Perplexity calculations **Coding:** Bigram model with smoothing
3	**Word embeddings**	Word2vec, GloVe, cosine similarity	Analogy and similarity calculations **Coding:** Train toy embeddings
4	**Text classification**	Bag-of-words, TF-IDF, logistic regression	TF-IDF by hand **Coding:** Sentiment classifier
5	**Sequence labelling**	HMMs, CRFs, Viterbi	Viterbi decoding **Coding:** Implement Viterbi
6	**Neural sequence models**	RNN language models, seq2seq	Sequence loss calculations **Coding:** Seq2seq toy task
7	**Attention and Transformers for text**	Encoder/decoder, masking	Attention masks **Coding:** Masked attention
8	**Pretrained encoders**	BERT, masked LM, fine-tuning	Fine-tuning parameter counts **Coding:** Fine-tune a classifier
9	**Parsing and semantics**	Dependency/constituency parsing, semantic roles	Parse trees **Coding:** Dependency parse features
10	**Evaluation**	BLEU, ROUGE, F1, human eval	Computing BLEU and F1 **Coding:** Implement metrics

Notes. Follows HKU COMP3361 / HKUST COMP4221.

Proof. Proofs will be added later.

QNT-308 · AI / ML

LLM / Generative AI

Stage 7
11 lectures x 3 h

Large language models: pretraining, scaling, alignment, prompting and retrieval.

Prerequisites	QNT-307 NLP
University courses	CUHK STAT6209
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Explain how LLMs are trained and aligned
- Use prompting, RAG and fine-tuning
- Evaluate risks and limits

Course outline

#	Topic	Theory	Computation / Coding
1	**Decoder-only Transformers**	Architecture, causal masking, KV cache	Parameter counts **Coding:** Implement a tiny GPT block
2	**Pretraining**	Objectives, data pipelines, tokenisation	Tokens-seen and cost estimates **Coding:** Streaming data loader
3	**Scaling laws**	Compute-optimal scaling	Scaling-law fits **Coding:** Fit a power law
4	**Sampling and decoding**	Temperature, top-k, top-p, beam	Probabilities under sampling **Coding:** Implement sampling
5	**Fine-tuning**	SFT, LoRA, parameter-efficient methods	LoRA rank and parameter counts **Coding:** LoRA layer
6	**Alignment**	RLHF, DPO, reward models	DPO loss by hand **Coding:** DPO loss
7	**Prompting and in-context learning**	Few-shot, chain of thought	Prompt design cases **Coding:** Prompt templates
8	**Retrieval-augmented generation**	Embeddings, vector search, chunking	Similarity ranking **Coding:** Minimal RAG pipeline
9	**Agents and tool use**	Function calling, planning	Tool-call traces **Coding:** Tool-calling loop
10	**Evaluation and safety**	Benchmarks, hallucination, red-teaming	Metric calculations **Coding:** Evaluate a model on a test set
11	**Efficiency**	Quantisation, distillation, serving	Memory estimates **Coding:** Quantise weights

Notes. Follows CUHK STAT6209 style, extended with current practice.

Proof. Proofs will be added later.

QNT-401 · Derivatives & Risk

Derivatives I

Stage 3
12 lectures x 3 h

Forwards, futures, options and no-arbitrage pricing in discrete time, up to Black-Scholes as a limit.

Prerequisites	QNT-201 Probability I
University courses	HKU FINA2322 · HKUST FINA3203
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Price forwards and options by no-arbitrage
- Use binomial trees
- Read option payoffs and put-call parity

Course outline

#	Topic	Theory	Computation / Coding
1	**Introduction to derivatives, forwards and options**	Forward and option payoffs, long and short, hedging vs speculation, the no-arbitrage principle, continuous compounding	Payoff and profit tables, compounding conversions, breakeven points
2	**Insurance, collars and other strategies**	Protective put, covered call, collar, spreads, straddle, butterfly	Combined payoffs, maximum profit and loss, breakeven points
3	**Introduction to risk management**	Why hedge, costs and limits of hedging	Hedged profit and loss, effective price
4	**Financial forwards and futures**	Cost of carry, with and without dividends, currency forwards, cash-and-carry arbitrage, margin and daily settlement	Forward price, arbitrage profit, margin account
5	**Commodity forwards and futures**	Storage costs, convenience yield, contango and backwardation	Commodity forward pricing, implied convenience yield
6	**Interest rate forwards and futures**	Forward rates, FRAs, Eurodollar and SOFR futures	Forward rates from a zero curve, FRA settlement
7	**Swaps**	Interest rate, currency and commodity swaps; a swap as a strip of forwards	Par swap rate, swap valuation
8	**Parity and other option relationships**	Put-call parity, price bounds, early exercise of American options	Checking parity, finding arbitrage
9	**Binomial option pricing**	Replicating portfolio, risk-neutral probability, multiple periods, American options, calibrating u and d from σ	One- and multi-period pricing, American option trees
10	**The Black-Scholes formula**	The formula (used, not derived), Greeks, implied volatility	Pricing with the formula, computing Greeks
11	**Market-making and delta-hedging**	Delta hedging, gamma risk, hedging P&L for a market maker	Delta-hedge P&L, delta-gamma approximation
12	**Financial engineering and applications**	Structured products, options in corporate finance (convertibles, warrants). HKU marks this as time permitting	Pricing a simple structured product

Notes. Follows the 13 topics of HKU FINA2322, merging only the first two. Black-Scholes is used but not derived; the derivation belongs to Derivatives II.

Proof. Proofs will be added later, for example cash-and-carry, put-call parity and replication.

QNT-402 · Derivatives & Risk

Credit Risk

Stage 4
10 lectures x 3 h

Credit risk measurement and pricing: default models, spreads and credit derivatives.

Prerequisites	QNT-401 Derivatives I, QNT-202 Probability II
University courses	HKU FINA3322 · CUHK RMSC6002 · HKUST MAFS5220
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Compute default probabilities and expected loss
- Price CDS and bonds with credit risk
- Model portfolio default correlation

Course outline

#	Topic	Theory	Computation / Coding
1	**Credit risk overview**	PD, LGD, EAD, expected loss	Expected loss calculations
2	**Credit ratings and transition matrices**	Markov rating models	Multi-year PDs
3	**Structural models**	Merton model, distance to default	Merton PD
4	**Reduced-form models**	Hazard rates, intensity	Survival probabilities
5	**Risky bond pricing**	Credit spreads, recovery	Spread decomposition
6	**Credit default swaps**	CDS valuation, bootstrapping	CDS spread calculations
7	**Portfolio credit and copulas**	Gaussian copula, correlation	Joint default probabilities
8	**CDOs**	Tranching, base correlation	Tranche losses
9	**Counterparty risk**	CVA, DVA, netting	CVA approximations
10	**Credit scoring**	Logistic scoring, validation	AUC and cut-off decisions

Notes. Follows HKU FINA3322 and CUHK RMSC6002.

Proof. Proofs will be added later.

QNT-403 · Derivatives & Risk

Interest Rate I

Stage 4
10 lectures x 3 h

Fixed income fundamentals: bonds, yields, duration and the term structure.

Prerequisites	QNT-401 Derivatives I
University courses	HKU FINA3323 · HKUST FINA4303
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Price bonds and swaps
- Compute duration and convexity
- Bootstrap curves

Course outline

#	Topic	Theory	Computation / Coding
1	**Time value and bond pricing**	PV, yield to maturity	Bond prices and yields
2	**Yield curves**	Spot, forward, par rates	Spot from par
3	**Bootstrapping**	Discount factors from instruments	Bootstrap a curve
4	**Duration**	Macaulay, modified, DV01	Duration and DV01
5	**Convexity**	Price-yield curvature	Price change approximations
6	**Immunisation and hedging**	Duration matching	Immunising liabilities
7	**Forward rate agreements and futures**	FRAs, Eurodollar futures, convexity adjustment	FRA pricing
8	**Interest rate swaps**	Fixed vs floating, valuation	Swap rates and values
9	**Term-structure theories**	Expectations, liquidity premium	Interpreting curves
10	**Bond options and caps**	Black-76 caps/floors	Cap prices

Notes. Follows HKU FINA3323 and HKUST FINA4303.

Proof. Proofs will be added later.

QNT-404 · Derivatives & Risk

Risk Management / VaR

Stage 4
11 lectures x 3 h

Measuring and managing market risk with VaR, expected shortfall and stress tests.

Prerequisites	QNT-502 Portfolio & Asset Pricing I, QNT-202 Probability II
University courses	HKU FINA4341 · CUHK RMSC5003
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Calculate VaR and ES under several methods
- Backtest risk models
- Design stress and scenario tests

Course outline

#	Topic	Theory	Computation / Coding
1	**Risk measures**	Volatility, VaR, ES, coherence	VaR from distributions
2	**Parametric VaR**	Variance-covariance, delta-normal	Portfolio VaR
3	**Historical simulation**	Empirical VaR, filtered HS	Empirical VaR
4	**Monte Carlo VaR**	Simulation of risk factors	Simulated quantiles
5	**Expected shortfall**	ES definition, estimation	ES calculations
6	**Volatility forecasting for risk**	EWMA, GARCH	EWMA recursions
7	**Backtesting**	Kupiec, Christoffersen tests	Exception counts and test stats
8	**Extreme value theory**	POT, GPD tails	Tail quantiles
9	**Stress testing**	Scenarios, reverse stress tests	Scenario P&L
10	**Portfolio risk attribution**	Component VaR, marginal risk	Risk contributions
11	**Liquidity and model risk**	Liquidity-adjusted VaR	Adjustments

Notes. Follows HKU FINA4341 and CUHK RMSC5003.

Proof. Proofs will be added later.

QNT-405 · Derivatives & Risk

Risk Theory & Insurance

Stage 4
10 lectures x 3 h

Risk theory: aggregate claims, ruin and premium principles.

Prerequisites	QNT-202 Probability II
University courses	CUHK RMSC4004 (course title only)
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Model claim frequency and severity
- Compute ruin probabilities
- Price insurance premiums

Course outline

#	Topic	Theory	Computation / Coding
1	**Utility and insurance**	Expected utility, risk aversion	Premium under utility
2	**Individual risk model**	Sums of independent risks	Distribution of total claims
3	**Collective risk model**	Compound Poisson	Aggregate claim moments
4	**Claim frequency**	Poisson, negative binomial, (a,b,0) class	Frequency fits
5	**Claim severity**	Pareto, lognormal, deductibles	Excess-of-loss calculations
6	**Premium principles**	Expected value, variance, Esscher	Premium loadings
7	**Reinsurance**	Proportional, stop-loss	Retained risk
8	**Ruin theory**	Cramer-Lundberg, Lundberg inequality	Ruin bounds
9	**Credibility theory**	Buhlmann credibility	Credibility estimates
10	**Generalised linear models in insurance**	Rating factors	Tariff calculations

Notes. Based on standard actuarial risk theory topics; CUHK RMSC4004 listed by title only.

Proof. Proofs will be added later.

QNT-406 · Derivatives & Risk

Operational Risk & Regulation

Stage 5
10 lectures x 3 h

Operational risk, regulatory capital and the Basel framework.

Prerequisites	QNT-404 Risk Management / VaR
University courses	CUHK RMSC4006 · CUHK RMSC3001 (Basel)
Assessment	Theory + Computation (Sufficient); Proof N/A, Complete = hardest set

Learning outcomes

- Compute regulatory capital
- Model loss distributions
- Understand the regulatory landscape

Course outline

#	Topic	Theory	Computation / Coding
1	**Basel framework overview**	Basel I-III, capital pillars	Capital ratio calculations
2	**Credit risk capital**	Standardised vs IRB	RWA computations
3	**Market risk capital**	Standardised approach, FRTB	Capital under FRTB
4	**Operational risk basics**	Event types, loss data	Classifying events
5	**Loss distribution approach**	Frequency-severity models	Aggregate loss simulation
6	**Extreme severity modelling**	Heavy tails, scenario analysis	Tail estimates
7	**Standardised measurement approach**	BIC, loss component	SMA capital
8	**Liquidity regulation**	LCR, NSFR	Ratio calculations
9	**Stress testing and ICAAP**	Supervisory stress tests	Capital projections
10	**Model governance**	Model risk, validation	Validation case study

Notes. Follows CUHK RMSC4006 and RMSC3001 (Basel). Course page only partly checked.

Proof. Proof = N/A may be considered; assessed through applied problems.

QNT-407 · Derivatives & Risk

Stochastic Calculus

Stage 5
10 lectures x 3 h

Ito calculus and its use in pricing: Brownian motion, SDEs, Girsanov and Black-Scholes.

Prerequisites	QNT-207 Stochastic Processes, QNT-106 Multivariable Calculus
University courses	HKU FINA3350 · CityU EF3520 · HKUST MAFS5010 · CUHK RMSC4005
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Compute with Ito's formula
- Solve and simulate SDEs
- Derive Black-Scholes by PDE and by change of measure

Course outline

#	Topic	Theory	Computation / Coding
1	**Brownian motion**	Definition, scaling, quadratic variation	Covariances of Brownian paths
2	**Properties of paths**	Non-differentiability, reflection principle	Hitting-time probabilities
3	**Ito integral**	Construction, isometry, martingale property	Moments of Ito integrals
4	**Ito's formula**	One- and multi-dimensional versions	Applying Ito to functions of BM
5	**Geometric Brownian motion**	Solution, lognormal distribution	GBM moments and probabilities
6	**SDEs**	Existence, OU process, linear SDEs	Solving linear SDEs
7	**Feynman-Kac**	PDE-expectation link	Solving PDEs via expectations
8	**Girsanov and measure change**	Radon-Nikodym, risk-neutral measure	Drift changes
9	**Black-Scholes derivation**	Self-financing replication, PDE	Verifying the BS PDE
10	**Martingale representation and numeraires**	Change of numeraire, forward measure	Computing prices under numeraires

Notes. Follows HKU FINA3350, HKUST MAFS5010 and CUHK RMSC4005. Builds on Stochastic Processes.

Proof. Proofs will be added later.

QNT-408 · Derivatives & Risk

Derivatives II

Stage 6
10 lectures x 3 h

Continuous-time derivative pricing beyond vanilla options: exotics, American options and hedging.

Prerequisites	QNT-401 Derivatives I, QNT-407 Stochastic Calculus
University courses	CityU EF4820 · HKUST MAFS5030
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Price options under Black-Scholes
- Handle exotics and early exercise
- Compute and manage Greeks

Course outline

#	Topic	Theory	Computation / Coding
1	**Black-Scholes revisited**	PDE and risk-neutral pricing	Closed-form call/put prices
2	**Greeks**	Delta, gamma, vega, theta, rho	Greek calculations
3	**Delta-gamma hedging**	Dynamic hedging, hedging error	Hedge P&L examples
4	**Barrier options**	Reflection, in-out parity	Barrier prices
5	**Asian and lookback options**	Average-price, lookbacks	Geometric Asian prices
6	**American options**	Optimal stopping, free boundary	Early-exercise comparisons
7	**Binomial and trinomial trees**	Convergence to BS	Tree prices
8	**Finite-difference methods**	Explicit, implicit, Crank-Nicolson	Stability checks
9	**Currency, dividend and commodity options**	Garman-Kohlhagen, futures options	Pricing variants
10	**Multi-asset options**	Exchange, basket, spread options	Margrabe and basket approximations

Notes. Follows HKUST MAFS5030 and CityU EF4820.

Proof. Proofs will be added later.

QNT-409 · Derivatives & Risk

Interest Rate II

Stage 6
10 lectures x 3 h

Stochastic models for interest rates and the pricing of rate derivatives.

Prerequisites	QNT-403 Interest Rate I, QNT-407 Stochastic Calculus
University courses	CityU EF4821 · HKUST MAFS5040 · CUHK RMSC6001
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Use short-rate and HJM models
- Price caps, floors and swaptions
- Apply the LIBOR market model

Course outline

#	Topic	Theory	Computation / Coding
1	**Short-rate models**	Vasicek, CIR, Hull-White	Bond prices in Vasicek
2	**Affine term structure**	Affine class, Riccati equations	Solving affine ODEs
3	**Calibration to the curve**	Hull-White fitting	Theta(t) fitting
4	**HJM framework**	Drift condition	HJM drift calculations
5	**Forward measures**	Change of numeraire for rates	Forward-measure pricing
6	**Caps, floors, swaptions**	Black formulas, annuity measure	Swaption prices
7	**LIBOR market model**	Lognormal forward rates, drift terms	LMM drift
8	**Calibration to vols**	Cap/swaption calibration	Calibration targets
9	**Multi-curve and OIS discounting**	Post-crisis curves, SOFR	Curve construction
10	**Numerical methods for rates**	Trees and Monte Carlo for rates	Simulation estimates

Notes. Follows HKUST MAFS5040 and CityU EF4821.

Proof. Proofs will be added later.

QNT-410 · Derivatives & Risk

Structured Products

Stage 7
10 lectures x 3 h

Pricing and risk of structured products built from vanillas and exotics.

Prerequisites	QNT-408 Derivatives II, QNT-204 Monte Carlo
University courses	HKU FINA4354 · HKUST MAFS5330
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Decompose a product into components
- Price with Monte Carlo and PDEs
- Identify the dominant risks

Course outline

#	Topic	Theory	Computation / Coding
1	**Structured product landscape**	Notes, payoffs, issuer credit	Decomposing payoffs
2	**Capital-protected notes**	Zero plus option replication	Participation rates
3	**Reverse convertibles**	Short put embedded	Coupon and breakeven
4	**Autocallables**	Barriers, early redemption	Autocall probabilities
5	**Pricing autocallables**	Monte Carlo and PDE	Simulated prices
6	**Worst-of products**	Correlation, basket payoffs	Correlation sensitivity
7	**Rate-linked products**	CMS, range accruals	Accrual calculations
8	**Hedging structured products**	Pin risk, barrier shifts	Hedge ratios
9	**Risk factors**	Vega, correlation, skew	Risk reports
10	**Regulation and suitability**	KID/PRIIPs, mis-selling	Suitability cases

Notes. Follows HKU FINA4354 and HKUST MAFS5330.

Proof. Proofs will be added later.

QNT-411 · Derivatives & Risk

Volatility Models

Stage 7
10 lectures x 3 h

Modelling the implied volatility surface with local, stochastic and parametric models.

Prerequisites	QNT-408 Derivatives II, QNT-107 Numerical Methods I
University courses	No university course (local vol, Heston, SABR)
Assessment	Theory + Computation (Sufficient); add Proof for Complete
Origin	Original to this app (no direct university course)

Learning outcomes

- Read and arbitrage-check a smile
- Calibrate local and stochastic volatility models
- Hedge with smile models

Course outline

#	Topic	Theory	Computation / Coding
1	**Implied volatility**	Quotes, Newton inversion	Implied vol from prices **Coding:** Implement an IV solver
2	**The smile and surface**	Skew, term structure, no-arbitrage conditions	Checking butterfly arbitrage **Coding:** Arbitrage checks
3	**Local volatility**	Dupire formula	Local vol from a surface **Coding:** Dupire on a grid
4	**Stochastic volatility**	Heston model, CIR variance	Moments of variance **Coding:** Simulate Heston
5	**Heston pricing**	Characteristic function, Fourier pricing	Characteristic function evaluation **Coding:** Fourier pricer
6	**SABR**	SABR model, Hagan expansion	SABR implied vol **Coding:** Implement SABR
7	**SVI and parametrisations**	SVI, no-arbitrage constraints	SVI fits **Coding:** Fit SVI
8	**Calibration**	Least squares, regularisation	Calibration error **Coding:** Calibrate to quotes
9	**Smile dynamics**	Sticky strike vs sticky delta	Hedge ratios **Coding:** Compare smile dynamics
10	**Exotics under smile models**	Local-stochastic mixtures	Model risk comparison **Coding:** Barrier price under two models

Notes. No university course; structure follows standard quant literature on smile modelling.

Proof. Proofs will be added later.

QNT-501 · Markets & Trading

Microstructure I

Stage 3
10 lectures x 3 h

How markets actually work: order books, spreads and price formation.

Prerequisites	QNT-201 Probability I
University courses	HKUST FINA4103 · CityU EF4323
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Read order books and trade data
- Decompose spreads
- Understand execution costs

Course outline

#	Topic	Theory	Computation / Coding
1	**Market structure**	Exchanges, dark pools, venues	Venue comparison
2	**Order types and the limit order book**	Limit, market, stop, iceberg	Book walk-through
3	**Matching engines**	Price-time priority, auctions	Matching examples
4	**Bid-ask spread**	Quoted, effective, realised spread	Spread calculations
5	**Spread components**	Order-processing, inventory, adverse selection	Roll estimator
6	**Price impact**	Kyle's lambda, square-root law	Impact estimates
7	**Information and trading**	Informed trading, PIN	Informativeness
8	**Transaction cost analysis**	Implementation shortfall, VWAP	Shortfall calculations
9	**Market quality**	Liquidity, depth, resilience	Liquidity metrics
10	**High-frequency trading**	Latency, co-location	Latency arbitrage cases

Notes. Follows HKUST FINA4103 and CityU EF4323.

Proof. Proofs will be added later.

QNT-502 · Markets & Trading

Portfolio & Asset Pricing I

Stage 3
10 lectures x 3 h

Mean-variance portfolio theory and the CAPM.

Prerequisites	QNT-201 Probability I, QNT-106 Multivariable Calculus
University courses	HKU FINA2320 · CityU EF3320
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Build efficient portfolios
- Apply the CAPM
- Evaluate portfolio performance

Course outline

#	Topic	Theory	Computation / Coding
1	**Returns and risk**	Simple vs log returns, mean, variance	Return statistics
2	**Portfolio variance**	Covariance, diversification	Two-asset portfolios
3	**Efficient frontier**	Minimum variance, frontier	Frontier weights
4	**Risk-free asset**	Capital allocation line, tangency portfolio	Tangency weights
5	**CAPM**	Beta, security market line	Expected returns from beta
6	**Beta estimation**	Market model	Beta from data
7	**Performance measures**	Sharpe, Treynor, Jensen alpha	Ratios
8	**Utility and risk aversion**	Mean-variance utility	Optimal allocations
9	**Constraints and estimation error**	Shorting limits, shrinkage	Constrained allocations
10	**Market efficiency**	EMH forms, anomalies	Event study calculation

Notes. Follows HKU FINA2320 and CityU EF3320.

Proof. Proofs will be added later.

QNT-503 · Markets & Trading

Market Making

Stage 4
10 lectures x 3 h

Quoting two-sided markets: spread setting, inventory, adverse selection and risk limits.

Prerequisites	QNT-501 Microstructure I, QNT-603 EV & Kelly, QNT-604 Game Theory
University courses	No university course
Assessment	Theory + Computation (Sufficient); add Proof for Complete
Origin	Original to this app (no direct university course)

Learning outcomes

- Set quotes from a fair value and edge
- Manage inventory and skew
- Judge adverse selection and size risk

Course outline

#	Topic	Theory	Computation / Coding
1	**What a market maker does**	Bid-ask spread, inventory, adverse selection, PnL decomposition, order book and queue priority	PnL and position from a sequence of fills
2	**Glosten-Milgrom model**	Informed vs uninformed traders, updating fair value from trade direction with Bayes	Computing the bid and the ask
3	**Inventory risk and quoting**	Avellaneda-Stoikov reservation price and optimal spread, inventory skew	Reservation price and spread from the formula
4	**Order flow and adverse selection**	Order flow toxicity, value of queue position, fill probability	Expected payoff of a queue position
5	**Fair value estimation**	Microprice, order book imbalance, combining signals	Microprice and imbalance signals
6	**Risk limits and sizing**	Position limits, volatility-scaled quotes, Kelly-style sizing, drawdown	Spread needed to reach a target Sharpe ratio
7	**Competition between market makers**	Penny-jumping, spread competition, queue priority games	Equilibrium spread in a simple game
8	**Making markets in binary contracts**	Prices between 0 and 1, resolution risk, fees, inventory management	Quoting a binary contract and its expected profit
9	**Backtest bias and performance evaluation**	Fill assumption bias, look-ahead bias, effect of latency and fees on PnL	Sharpe ratio, t-statistic of edge per trade, PnL net of adverse selection
10	**Full strategy design**	Designing a strategy end to end, PnL attribution, fill rate, inventory half-life	PnL attribution calculations

Notes. No university course; designed from scratch. No coding: theory, computation and proof only.

Proof. Proof candidates: Glosten-Milgrom bid and ask as Bayesian updates (lecture 2), the Avellaneda-Stoikov reservation price (lecture 3), Kelly position sizing (lecture 6), the equilibrium spread in a simple game (lecture 7), the fair value of a binary contract (lecture 8).

QNT-504 · Markets & Trading

Options & Vol Trading

Stage 4
11 lectures x 3 h

Trading options and volatility: Greeks, vol surfaces, spreads and risk management.

Prerequisites	QNT-401 Derivatives I, QNT-603 EV & Kelly
University courses	CUHK FINA4370 (course title only)
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Trade volatility views with options
- Manage Greeks as a book
- Evaluate P&L attribution

Course outline

#	Topic	Theory	Computation / Coding
1	**Option market basics**	Chains, quoting, conventions	Reading chains
2	**Greeks in practice**	Delta, gamma, vega, theta, higher Greeks	Position Greeks
3	**Spreads and combinations**	Straddles, strangles, butterflies	Payoff diagrams
4	**Volatility as an asset**	Implied vs realised, variance swaps	Vol P&L
5	**Gamma-theta trade-off**	Breakeven moves	Breakeven calculations
6	**Delta hedging P&L**	Hedging frequency	Hedge P&L
7	**Skew and term structure trades**	Risk reversals, calendars	Skew trade P&L
8	**Event volatility**	Earnings, jumps	Event vol decomposition
9	**Dispersion and correlation**	Index vs single stock	Dispersion trades
10	**Portfolio risk**	Greek ladders, scenarios	Scenario P&L
11	**P&L attribution**	Greeks-based explain	Attribution

Notes. CUHK FINA4370 listed by title only; structure follows desk practice.

Proof. Proofs will be added later.

QNT-505 · Markets & Trading

Portfolio & Asset Pricing II

Stage 4
10 lectures x 3 h

Multi-factor models, APT and modern portfolio construction.

Prerequisites	QNT-502 Portfolio & Asset Pricing I, QNT-110 Optimization, QNT-203 Regression I
University courses	HKUST MAFS5210 · CUHK RMSC6006
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Estimate factor models
- Construct portfolios with risk models
- Test asset-pricing models

Course outline

#	Topic	Theory	Computation / Coding
1	**Multi-factor models**	APT, factor exposures	Factor-implied returns
2	**Fama-French factors**	Size, value, momentum, profitability	Factor return construction
3	**Time-series and cross-sectional tests**	Fama-MacBeth, GRS	Risk premium estimates
4	**Stochastic discount factor**	SDF, no-arbitrage pricing	SDF-implied prices
5	**Covariance estimation**	Shrinkage, factor covariance	Risk model covariances
6	**Black-Litterman**	Priors and views	Posterior returns
7	**Risk parity and robust optimisation**	Equal risk contribution	ERC weights
8	**Transaction costs and turnover**	Cost-aware optimisation	Net returns
9	**Factor timing and crowding**	Factor decay	Decay analysis
10	**Performance attribution**	Brinson, factor attribution	Attribution tables

Notes. Follows HKUST MAFS5210 and CUHK RMSC6006.

Proof. Proofs will be added later.

QNT-506 · Markets & Trading

Microstructure II

Stage 5
10 lectures x 3 h

Mathematical models of market microstructure, optimal execution and market making.

Prerequisites	QNT-501 Microstructure I, QNT-207 Stochastic Processes, QNT-110 Optimization
University courses	HKUST MAFS5270
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Solve optimal execution problems
- Model order flow
- Derive market-making quotes

Course outline

#	Topic	Theory	Computation / Coding
1	**Kyle model**	Single-period equilibrium	Lambda derivation
2	**Glosten-Milgrom**	Sequential trade, Bayesian updating	Quote updates
3	**Order-flow modelling**	Hawkes processes	Intensity calculations
4	**Limit-order-book models**	Queue models, Markov LOB	Fill probabilities
5	**Optimal execution I**	Almgren-Chriss	Optimal schedules
6	**Optimal execution II**	HJB, stochastic control	HJB solutions
7	**Market-making models**	Avellaneda-Stoikov	Optimal quotes
8	**Inventory control**	Risk aversion, reservation price	Reservation prices
9	**Impact and alpha**	Transient impact, propagator	Impact kernels
10	**Estimation**	Parameter calibration	Fit execution models

Notes. Follows HKUST MAFS5270 style.

Proof. Proofs will be added later.

QNT-507 · Markets & Trading

Algo / Quant Trading

Stage 6
11 lectures x 3 h

Designing, testing and running systematic trading strategies.

Prerequisites	QNT-203 Regression I, QNT-214 Time Series I, QNT-705 Python, QNT-501 Microstructure I
University courses	HKU FINA4359 · HKU STAT8020 · HKUST FINA4803
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Implement signal research and backtests
- Evaluate strategies with honest statistics
- Manage execution, risk and capacity

Course outline

#	Topic	Theory	Computation / Coding
1	**Strategy lifecycle**	Alpha, signals, portfolios, execution	Strategy anatomy **Coding:** Load price data
2	**Signal construction**	Momentum, mean reversion, carry	Signal calculations **Coding:** Compute signals
3	**Statistical arbitrage**	Pairs, cointegration	Spread z-scores **Coding:** Pairs strategy
4	**Factor and cross-sectional strategies**	Ranking, long-short	Portfolio returns **Coding:** Long-short backtest
5	**Backtesting engine**	Event-driven vs vectorised	Biases **Coding:** Build a backtester
6	**Transaction costs and slippage**	Cost models	Net P&L **Coding:** Cost model
7	**Performance evaluation**	Sharpe, drawdown, turnover	Metrics **Coding:** Performance report
8	**Overfitting control**	Walk-forward, deflated Sharpe	Adjusted Sharpe **Coding:** Walk-forward test
9	**Portfolio construction and sizing**	Vol targeting, Kelly	Position sizing **Coding:** Vol-targeted sizing
10	**Execution algorithms**	TWAP, VWAP, POV	Schedule calculations **Coding:** TWAP/VWAP
11	**Live trading and monitoring**	Risk limits, kill switches	Limits **Coding:** Monitoring checks

Notes. Follows HKU FINA4359, HKU STAT8020 and HKUST FINA4803, with coding emphasised.

Proof. Proofs will be added later.

QNT-601 · Problem Solving

Fermi Estimation

Stage 1
10 lectures x 3 h

Estimating unfamiliar quantities from anchors and structure.

Prerequisites	None
University courses	No university course
Assessment	Theory + Computation (Sufficient); Proof N/A, Complete = hardest set
Origin	Original to this app (no direct university course)

Learning outcomes

- Decompose estimation problems
- Choose good anchors
- Bound and sanity-check answers

Course outline

#	Topic	Theory	Computation / Coding
1	**Orders of magnitude**	Powers of 10, anchors	Magnitude estimates
2	**Decomposition**	Breaking into factors	Decomposition problems
3	**Population and market sizing**	Demographic anchors	Market size estimates
4	**Physical estimates**	Energy, mass, distance	Physical estimates
5	**Financial estimates**	Revenue, volumes, tick counts	Financial estimates
6	**Error and bounds**	Log-error propagation	Bound calculations
7	**Sanity checks**	Cross-checking	Consistency checks
8	**Trading-style estimation**	Volume, spreads, flows	Trading estimates
9	**Calibration**	Confidence intervals, overconfidence	Calibration quizzes
10	**Mixed timed sets**	Speed and accuracy	Timed sets

Notes. No university course; practice-driven.

Proof. Proof = N/A; Complete = pass the hardest set.

QNT-602 · Problem Solving

Mental Math

Stage 1
10 lectures x 3 h

Fast, accurate mental arithmetic.

Prerequisites	None
University courses	No university course
Assessment	Theory + Computation (Sufficient); Proof N/A, Complete = hardest set
Origin	Original to this app (no direct university course)

Learning outcomes

- Compute sums, products and fractions rapidly
- Estimate with controlled error
- Reach target speed and accuracy

Course outline

#	Topic	Theory	Computation / Coding
1	**Addition and subtraction**	Complements, chunking	Speed drills
2	**Multiplication I**	Distributive, doubling/halving	2x2-digit drills
3	**Multiplication II**	Squares, near-base tricks	Squaring drills
4	**Division and fractions**	Reciprocals, decimal equivalents	Fraction drills
5	**Percentages and ratios**	Percent shortcuts	Percent drills
6	**Estimation**	Rounding, error bounds	Estimation drills
7	**Powers and roots**	Squares/cubes tables, root estimation	Root drills
8	**Logs and exponentials**	Rule of 72, ln approximations	Log drills
9	**Probability arithmetic**	Fast binomials	Quick probability sums
10	**Mixed timed sets**	Speed under pressure	Timed sets

Notes. No university course; a drill-based topic.

Proof. Proof = N/A; Complete = pass the hardest speed set.

QNT-603 · Problem Solving

EV & Kelly

Stage 3
10 lectures x 3 h

Making decisions under uncertainty by expected value and optimal bet sizing.

Prerequisites	QNT-201 Probability I
University courses	No university course
Assessment	Theory + Computation (Sufficient); add Proof for Complete
Origin	Original to this app (no direct university course)

Learning outcomes

- Evaluate bets by EV and variance
- Size positions with Kelly
- Understand drawdown and ruin

Course outline

#	Topic	Theory	Computation / Coding
1	**Expected value basics**	EV of bets and games	Fair price of games
2	**Utility and risk preference**	Risk aversion, certainty equivalent	Certainty equivalents
3	**The Kelly criterion**	Log utility, growth rate	Optimal fraction
4	**Kelly with multiple outcomes**	Multi-outcome Kelly	Optimal allocations
5	**Fractional Kelly**	Risk-growth trade-off	Growth rate at fractions
6	**Continuous Kelly**	Gaussian returns, leverage	Optimal leverage
7	**Drawdowns and ruin**	Probability of drawdown	Drawdown probabilities
8	**Betting with edge uncertainty**	Estimation error	Shrunken Kelly
9	**Auctions and pricing games**	Winner's curse	Optimal bids
10	**Casino and market-style games**	Interview games	Game EV problems

Notes. No university course; draws on Kelly's paper and Thorp's work, as used in trading interviews.

Proof. Proofs will be added later.

QNT-604 · Problem Solving

Game Theory

Stage 3
10 lectures x 3 h

Strategic interaction: equilibria, auctions and information games.

Prerequisites	QNT-201 Probability I
University courses	HKU ECON2214 · CUHK ECON3160 · CityU EF4484
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Find Nash equilibria
- Analyse sequential and Bayesian games
- Apply auction theory

Course outline

#	Topic	Theory	Computation / Coding
1	**Normal-form games**	Dominance, best response	Dominance elimination
2	**Nash equilibrium**	Pure and mixed equilibria	Solving 2x2 mixed equilibria
3	**Zero-sum games**	Minimax theorem	Game values
4	**Extensive-form games**	Backward induction, subgame perfection	Solving game trees
5	**Repeated games**	Folk theorem, trigger strategies	Discounted payoffs
6	**Bayesian games**	Types, Bayesian Nash	Bayesian equilibria
7	**Auctions I**	First- and second-price	Equilibrium bids
8	**Auctions II**	Revenue equivalence	Expected revenue
9	**Signalling and screening**	Separating/pooling equilibria	Signalling equilibria
10	**Cooperative games and bargaining**	Shapley value, Nash bargaining	Shapley values

Notes. Follows HKU ECON2214 and CUHK ECON3160.

Proof. Proofs will be added later.

QNT-605 · Problem Solving

Probability Puzzles

Stage 3
10 lectures x 3 h

Interview-style probability and combinatorics puzzles solved with clean reasoning.

Prerequisites	QNT-201 Probability I, QNT-102 Discrete Math & Combinatorics
University courses	No university course
Assessment	Theory + Computation (Sufficient); add Proof for Complete
Origin	Original to this app (no direct university course)

Learning outcomes

- Set up problems quickly
- Use symmetry, recursion and conditioning
- Solve within interview time limits

Course outline

#	Topic	Theory	Computation / Coding
1	**Counting tricks**	Bijections, stars and bars, symmetry	Quick counting problems
2	**Conditioning and recursion**	First-step analysis	Expected values via recursion
3	**Dice and coin problems**	Waiting times, runs	Expected rolls and flips
4	**Cards and urns**	Hypergeometric, exchangeability	Card probabilities
5	**Random walks and ruin**	Reflection, gambler's ruin	Hitting probabilities
6	**Order statistics and geometric probability**	Uniform spacings, Buffon	Geometric probabilities
7	**Markov chain puzzles**	Absorbing states	Hitting time puzzles
8	**Indicator variables and linearity**	Linearity of expectation	Fixed points, coupon collector
9	**Bayes and information puzzles**	Monty Hall variants	Posterior calculations
10	**Timed mixed sets**	Speed and accuracy	Timed problem sets

Notes. No university course; follows the style of quant interview question banks.

Proof. Proofs will be added later.

QNT-701 · Computing

C++ I

Stage 1
11 lectures x 3 h

C++ fundamentals: types, memory and object-oriented programming.

Prerequisites	None
University courses	HKUST COMP2011
Assessment	Theory + Computation (Sufficient); Proof N/A, Complete = hardest set

Learning outcomes

- Write correct C++ programs
- Manage memory safely
- Use classes and the STL

Course outline

#	Topic	Theory	Computation / Coding
1	**Types, expressions, control flow**	Syntax, operators	Program tracing **Coding:** Basic programs
2	**Functions and references**	Pass by value/reference	Call traces **Coding:** Function tasks
3	**Arrays and pointers**	Pointer arithmetic	Memory diagrams **Coding:** Pointer tasks
4	**Dynamic memory**	new/delete, leaks	Lifetime analysis **Coding:** Memory tasks
5	**Classes and objects**	Constructors, encapsulation	Object tracing **Coding:** Class design
6	**Inheritance and polymorphism**	Virtual functions	Dispatch tracing **Coding:** Polymorphism tasks
7	**Operator overloading**	Overloads, copy semantics	Overload semantics **Coding:** Overload tasks
8	**Templates**	Function and class templates	Template deduction **Coding:** Template tasks
9	**STL containers**	vector, map, set	Container operations **Coding:** STL tasks
10	**STL algorithms**	Iterators, lambdas	Algorithm outputs **Coding:** STL algorithms
11	**Error handling and I/O**	Exceptions, streams	Exception flow **Coding:** I/O tasks

Notes. Follows HKUST COMP2011. Coding in C++ is planned for a later release.

Proof. Proof = N/A; Complete = pass the hardest set.

QNT-702 · Computing

Computer Architecture

Stage 1
12 lectures x 3 h

How computers execute programs: data representation, assembly, pipelines and memory.

Prerequisites	None
University courses	HKU COMP2120
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Represent and manipulate data in binary
- Read assembly
- Reason about caches and pipelines

Course outline

#	Topic	Theory	Computation / Coding
1	**Number systems**	Binary, hex, two's complement	Conversions
2	**Floating point**	IEEE 754	Float encodings
3	**Boolean logic**	Gates, combinational circuits	Truth tables
4	**Sequential logic**	Latches, registers, state machines	Timing diagrams
5	**Instruction set architecture**	RISC/CISC, registers	Instruction encoding
6	**Assembly programming**	Control flow, procedures	Assembly traces
7	**Processor datapath**	Single-cycle design	Datapath tracing
8	**Pipelining**	Hazards, forwarding	CPI calculations
9	**Memory hierarchy**	Locality, caches	Hit/miss analysis
10	**Cache design**	Associativity, replacement	Cache simulations
11	**Virtual memory**	Page tables, TLB	Address translation
12	**I/O and parallelism**	Interrupts, multicore	Speedup calculations

Notes. Follows HKU COMP2120.

Proof. Proofs will be added later.

QNT-703 · Computing

Dev Tools (Linux · Git)

Stage 1
10 lectures x 3 h

The command line, Git and everyday developer tooling.

Prerequisites	None
University courses	No standalone university course
Assessment	Theory + Computation (Sufficient); Proof N/A, Complete = hardest set
Origin	Original to this app (no direct university course)

Learning outcomes

- Work fluently in a Linux shell
- Use Git for solo and team work
- Automate routine tasks

Course outline

#	Topic	Theory	Computation / Coding
1	**Shell basics**	Files, paths, permissions	Command exercises **Coding:** Shell commands
2	**Text tools**	grep, sed, awk, pipes	Pipeline puzzles **Coding:** Text processing
3	**Shell scripting**	Variables, loops, functions	Script reading **Coding:** Write a script
4	**Processes and environment**	Signals, jobs, env vars	Process questions **Coding:** Process control
5	**Git basics**	Commits, staging, history	Repository states **Coding:** Git exercises
6	**Git branching**	Branch, merge, rebase	Merge outcomes **Coding:** Branch tasks
7	**Collaboration**	Remotes, pull requests, conflicts	Conflict resolution **Coding:** Resolve conflicts
8	**SSH and remote work**	Keys, tunnelling, tmux	Config reading **Coding:** SSH config
9	**Packaging and environments**	venv, pip, containers	Dependency problems **Coding:** Environment setup
10	**Build tools and automation**	Make, CI basics	Makefile reading **Coding:** Write a Makefile

Notes. No standalone university course; follows common developer curricula.

Proof. Proof = N/A; Complete = pass the hardest set.

QNT-704 · Computing

LaTeX & Research Writing

Stage 1
10 lectures x 3 h

Typesetting mathematics and writing research papers in LaTeX.

Prerequisites	None
University courses	No university course (math typesetting, figures, BibTeX citations, paper structure)
Assessment	Theory + Computation (Sufficient); Proof N/A, Complete = hardest set
Origin	Original to this app (no direct university course)

Learning outcomes

- Typeset equations and documents
- Manage figures and citations
- Structure a research paper

Course outline

#	Topic	Theory	Computation / Coding
1	**Document structure**	Classes, packages, sections	Compile questions **Coding:** Write a document
2	**Mathematical typesetting I**	Inline/display math, symbols	Equation fixes **Coding:** Typeset formulas
3	**Mathematical typesetting II**	align, matrices, theorem environments	Environment choices **Coding:** Aligned derivations
4	**Tables and figures**	tabular, booktabs, includegraphics	Float placement **Coding:** Figures and tables
5	**Plotting**	TikZ, pgfplots, matplotlib export	Plot specs **Coding:** Make a plot
6	**Citations**	BibTeX/BibLaTeX, styles	Bib entries **Coding:** Bibliography
7	**Cross-references and macros**	Labels, custom commands	Macro design **Coding:** Macros
8	**Paper structure**	Abstract, intro, related work	Structure review **Coding:** Outline a paper
9	**Slides and posters**	Beamer	Slide layout **Coding:** Make slides
10	**Collaboration**	Overleaf, Git for LaTeX	Workflow design **Coding:** Version-controlled paper

Notes. No university course; follows a standard LaTeX workflow for research papers.

Proof. Proof = N/A; Complete = pass the hardest set.

QNT-705 · Computing

Python

Stage 1
11 lectures x 3 h

Python programming from basics to numerical computing.

Prerequisites	None
University courses	HKU COMP1117 · HKUST MAFS5360
Assessment	Theory + Computation (Sufficient); Proof N/A, Complete = hardest set

Learning outcomes

- Write clear Python programs
- Use NumPy and pandas
- Debug and test code

Course outline

#	Topic	Theory	Computation / Coding
1	**Syntax and types**	Variables, types, operators	Expression evaluation **Coding:** Basic programs
2	**Control flow**	Conditionals, loops	Trace outputs **Coding:** Loop programs
3	**Functions**	Arguments, scope, recursion	Trace calls **Coding:** Write functions
4	**Data structures**	Lists, dicts, sets, tuples	Operation outputs **Coding:** Collection tasks
5	**Strings and files**	Parsing, text and CSV I/O	Parsing outputs **Coding:** Parse files
6	**Object-oriented programming**	Classes, inheritance	Class behaviours **Coding:** Define classes
7	**Errors and testing**	Exceptions, unit tests	Exception flow **Coding:** Write tests
8	**NumPy**	Arrays, broadcasting	Shape questions **Coding:** Vectorise code
9	**pandas**	DataFrames, groupby, joins	Query outputs **Coding:** Data wrangling
10	**Plotting and performance**	matplotlib, profiling	Complexity questions **Coding:** Plot and optimise
11	**Idiomatic Python**	Comprehensions, generators	Output tracing **Coding:** Refactor code

Notes. Follows HKU COMP1117 style, extended with numerical libraries.

Proof. Proof = N/A; Complete = pass the hardest coding set.

QNT-706 · Computing

DSA I

Stage 2
12 lectures x 3 h

Core data structures and algorithm analysis.

Prerequisites	QNT-705 Python, QNT-102 Discrete Math & Combinatorics
University courses	HKU COMP2119 · CUHK CSCI2100
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Analyse running time
- Implement core data structures
- Solve standard algorithm problems

Course outline

#	Topic	Theory	Computation / Coding
1	**Complexity analysis**	Big-O, recurrences	Solving recurrences **Coding:** Complexity-aware solutions
2	**Arrays, lists, stacks, queues**	Abstract data types	Operation traces **Coding:** Stack and queue tasks
3	**Hashing**	Hash tables, collisions	Hash table states **Coding:** Hash map problems
4	**Trees**	Traversals, BSTs	Tree traversals **Coding:** Tree problems
5	**Balanced trees**	AVL, red-black	Rotations **Coding:** Balanced BST
6	**Heaps and priority queues**	Heap operations, heapsort	Heap states **Coding:** Heap problems
7	**Sorting**	Merge, quick, lower bounds	Sorting traces **Coding:** Sorting tasks
8	**Searching and selection**	Binary search, quickselect	Search traces **Coding:** Binary search problems
9	**Graphs**	Representation, BFS, DFS	Traversal orders **Coding:** Graph traversal
10	**Shortest paths**	Dijkstra, Bellman-Ford	Distance tables **Coding:** Shortest path
11	**Minimum spanning trees**	Kruskal, Prim, union-find	MST by hand **Coding:** MST
12	**Divide and conquer, greedy**	Design paradigms	Algorithm choices **Coding:** Mixed problems

Notes. Follows HKU COMP2119 and CUHK CSCI2100, with LeetCode-style practice.

Proof. Proofs will be added later.

QNT-707 · Computing

C++ II

Stage 3
10 lectures x 3 h

Modern C++ for performance-critical numerical and trading code.

Prerequisites	QNT-701 C++ I, QNT-706 DSA I
University courses	HKUST MAFS5240 · HKU FITE7406
Assessment	Theory + Computation (Sufficient); Proof N/A, Complete = hardest set

Learning outcomes

- Use modern C++ idioms
- Write efficient, safe code
- Implement numerical pricing code

Course outline

#	Topic	Theory	Computation / Coding
1	**Move semantics**	Rvalues, move constructors	Ownership reasoning **Coding:** Move tasks
2	**Smart pointers and RAII**	unique_ptr, shared_ptr	Lifetime analysis **Coding:** RAII tasks
3	**Modern language features**	auto, constexpr, structured bindings	Feature usage **Coding:** Modern syntax tasks
4	**Advanced templates**	SFINAE, concepts, CRTP	Template questions **Coding:** Template design
5	**Concurrency I**	Threads, mutexes	Race analysis **Coding:** Threaded tasks
6	**Concurrency II**	Atomics, memory model	Ordering questions **Coding:** Atomic tasks
7	**Performance**	Cache behaviour, profiling	Cost estimates **Coding:** Optimisation tasks
8	**Numerical libraries**	Eigen, BLAS	Complexity of kernels **Coding:** Matrix tasks
9	**Pricing library design**	Option pricer with polymorphism	Design review **Coding:** Pricer implementation
10	**Testing and build systems**	CMake, unit tests	Build reading **Coding:** Test and build

Notes. Follows HKUST MAFS5240 and HKU FITE7406 style (modern C++ for finance).

Proof. Proof = N/A; Complete = pass the hardest set.

QNT-708 · Computing

Computer Networks

Stage 3
11 lectures x 3 h

How data moves across networks: protocols, routing and transport.

Prerequisites	QNT-706 DSA I
University courses	HKU COMP3234
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Explain the layered model
- Analyse TCP and routing
- Reason about latency and throughput

Course outline

#	Topic	Theory	Computation / Coding
1	**Network overview**	Layers, packet switching	Delay calculations
2	**Application layer**	HTTP, DNS	Protocol traces
3	**Sockets**	Client/server sockets	Socket programs
4	**Transport I**	UDP, reliability	Retransmission analysis
5	**Transport II**	TCP, flow control	Window dynamics
6	**Congestion control**	Slow start, AIMD	Congestion windows
7	**Network layer**	IP, addressing, NAT	Subnet calculations
8	**Routing**	Link-state, distance-vector	Routing tables
9	**Link layer**	Ethernet, ARP, switches	Frame handling
10	**Wireless and multicast**	WiFi basics	Capacity estimates
11	**Network security**	TLS, firewalls	Handshake analysis

Notes. Follows HKU COMP3234.

Proof. Proofs will be added later.

QNT-709 · Computing

DSA II

Stage 3
11 lectures x 3 h

Advanced algorithm design: dynamic programming, flows, NP-completeness and approximation.

Prerequisites	QNT-706 DSA I
University courses	CUHK CSCI3160 · CityU CS4335
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Design DP and greedy algorithms
- Model problems as flows
- Prove hardness and design approximations

Course outline

#	Topic	Theory	Computation / Coding
1	**Dynamic programming I**	Optimal substructure, memoisation	DP tables **Coding:** DP problems
2	**Dynamic programming II**	Knapsack, sequence alignment	DP tables **Coding:** DP problems
3	**Greedy algorithms**	Exchange arguments, matroids	Greedy choices **Coding:** Greedy problems
4	**Network flow**	Max-flow min-cut, Ford-Fulkerson	Residual graphs **Coding:** Flow problems
5	**Matching and flow applications**	Bipartite matching, assignment	Matchings **Coding:** Matching problems
6	**Linear programming**	Duality, simplex view	LP formulations **Coding:** LP modelling
7	**Amortised analysis**	Aggregate, potential methods	Amortised costs **Coding:** Dynamic arrays
8	**String algorithms**	KMP, suffix arrays	Pattern matching **Coding:** String problems
9	**NP-completeness**	Reductions, SAT	Reduction steps **Coding:** Reduction tasks
10	**Approximation algorithms**	Vertex cover, TSP	Approximation ratios **Coding:** Approximation tasks
11	**Randomised algorithms**	Quicksort, hashing, Monte Carlo	Expected costs **Coding:** Randomised problems

Notes. Follows CUHK CSCI3160 and CityU CS4335.

Proof. Proofs are central here, so Proof carries more weight.

QNT-710 · Computing

Databases

Stage 3
11 lectures x 3 h

Relational databases, SQL and the basics of transactions and storage.

Prerequisites	QNT-706 DSA I
University courses	CUHK CSCI3170 · CityU CS3402
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Design relational schemas
- Write correct SQL
- Reason about transactions and indexes

Course outline

#	Topic	Theory	Computation / Coding
1	**Relational model**	Keys, relational algebra	Algebra expressions **Coding:** Basic SELECT queries
2	**SQL I**	SELECT, WHERE, joins	Query results **Coding:** Join queries
3	**SQL II**	Aggregation, subqueries, windows	Query results **Coding:** Aggregate and window queries
4	**ER modelling**	Entities, relationships	ER diagrams **Coding:** Schema from ER
5	**Normalisation**	Functional dependencies, BCNF	Normal-form decomposition **Coding:** Normalise a schema
6	**Indexing**	B+ trees, hashing	Index costs **Coding:** Index design
7	**Query processing**	Cost-based optimisation	Plan costs **Coding:** Read query plans
8	**Transactions**	ACID, isolation levels	Anomaly analysis **Coding:** Transaction cases
9	**Concurrency control**	Locking, MVCC	Schedule serialisability **Coding:** Lock reasoning
10	**Recovery**	WAL, checkpoints	Log analysis **Coding:** Recovery cases
11	**NoSQL and time-series stores**	Key-value, column stores	Model choices **Coding:** Market-data schemas

Notes. Follows CUHK CSCI3170 and CityU CS3402, with SQL judged by hidden tests.

Proof. Proofs will be added later.

QNT-711 · Computing

Operating Systems

Stage 3
11 lectures x 3 h

How operating systems manage processes, memory and storage.

Prerequisites	QNT-701 C++ I, QNT-706 DSA I, QNT-702 Computer Architecture
University courses	HKU COMP3230 · CUHK CSCI3150
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Reason about scheduling and concurrency
- Understand virtual memory
- Analyse file systems

Course outline

#	Topic	Theory	Computation / Coding
1	**OS overview**	Kernel, system calls	System call traces
2	**Processes**	Creation, context switch	Process tree analysis
3	**Threads**	Threading models, pthreads	Thread interleavings
4	**CPU scheduling**	FCFS, SJF, RR, MLFQ	Gantt charts, waiting times
5	**Synchronisation**	Locks, semaphores, monitors	Race analysis
6	**Deadlocks**	Conditions, banker's algorithm	Deadlock detection
7	**Memory management**	Paging, segmentation	Address translation
8	**Virtual memory**	Page replacement, thrashing	Page-fault counts
9	**File systems**	Inodes, journalling	Block allocation
10	**I/O and storage**	Disk scheduling, SSD	Seek calculations
11	**Security and virtualisation**	Isolation, VMs, containers	Isolation scenarios

Notes. Follows HKU COMP3230 and CUHK CSCI3150.

Proof. Proofs will be added later.

QNT-712 · Computing

Distributed Systems

Stage 4
11 lectures x 3 h

Building systems that run on many machines and survive failures.

Prerequisites	QNT-711 Operating Systems, QNT-708 Computer Networks
University courses	HKU COMP3358 · CUHK CSCI4160 · HKU COMP7305
Assessment	Theory + Computation (Sufficient); add Proof for Complete

Learning outcomes

- Reason about consistency and failure
- Understand consensus
- Evaluate distributed design trade-offs

Course outline

#	Topic	Theory	Computation / Coding
1	**Distributed systems basics**	Failures, models	Failure scenarios
2	**Time and ordering**	Logical and vector clocks	Clock calculations
3	**Remote procedure calls**	RPC semantics	Call semantics
4	**Replication**	Primary-backup, quorums	Quorum sizes
5	**Consistency models**	Linearisability, eventual consistency	Consistency analysis
6	**Consensus I**	Paxos	Paxos traces
7	**Consensus II**	Raft	Raft traces
8	**CAP and PACELC**	Trade-offs	Classifying systems
9	**Distributed storage**	Sharding, consistent hashing	Hash ring placement
10	**Distributed computation**	MapReduce, stream processing	Dataflow reasoning
11	**Fault tolerance**	Checkpointing, Byzantine faults	Failure probabilities

Notes. Follows HKU COMP3358 and CUHK CSCI4160.

Proof. Proofs will be added later.

QNT-713 · Computing

Low-latency Programming

Stage 4
11 lectures x 3 h

Writing code that runs in microseconds: memory, cache, lock-free structures and measurement.

Prerequisites	QNT-707 C++ II, QNT-711 Operating Systems
University courses	No university course
Assessment	Theory + Computation (Sufficient); Proof N/A, Complete = hardest set
Origin	Original to this app (no direct university course)

Learning outcomes

- Profile and measure latency
- Design cache-friendly and lock-free code
- Understand the trading hot path

Course outline

#	Topic	Theory	Computation / Coding
1	**Latency fundamentals**	Tail latency, percentiles	Percentile calculations
2	**Measuring performance**	Benchmarking, perf tools	Benchmark interpretation
3	**CPU and cache**	Cache lines, false sharing	Cache miss counts
4	**Memory management**	Allocators, pools, NUMA	Allocation costs
5	**Data layout**	SoA vs AoS, alignment	Layout analysis
6	**Branching and compilation**	Branch prediction, inlining	Assembly reading
7	**Lock-free programming**	Atomics, ring buffers	Memory-order reasoning
8	**Kernel bypass**	DPDK, busy polling	Latency budgets
9	**Network and market data**	UDP multicast, feed handlers	Feed decoding
10	**Order gateway design**	Hot path layout	Hot-path review
11	**Production practice**	Monitoring, determinism	Jitter analysis

Notes. No university course; follows low-latency trading engineering practice.

Proof. Proof = N/A; Complete = pass the hardest set.

QNT-714 · Computing

Software Engineering & System Design

Stage 4
11 lectures x 3 h

Building maintainable software and designing scalable systems.

Prerequisites	QNT-706 DSA I, QNT-710 Databases
University courses	HKU COMP3297
Assessment	Theory + Computation (Sufficient); Proof N/A, Complete = hardest set

Learning outcomes

- Apply software engineering practice
- Design scalable systems
- Reason about trade-offs

Course outline

#	Topic	Theory	Computation / Coding
1	**Software process**	Agile, requirements	Requirement analysis
2	**Design principles**	SOLID, modularity	Design critique
3	**Design patterns**	Creational, structural, behavioural	Pattern selection
4	**Testing**	Unit, integration, TDD	Test design
5	**APIs and services**	REST, versioning	API design
6	**System design fundamentals**	Latency, throughput, SLAs	Capacity estimation
7	**Storage and caching**	Sharding, caching	Cache hit-rate estimates
8	**Messaging and queues**	Pub/sub, streams	Throughput estimates
9	**Reliability**	Redundancy, failure modes	Availability calculations
10	**Case study: trading system**	Order management design	Design walkthrough
11	**Case study: data platform**	Market-data pipeline	Design walkthrough

Notes. Follows HKU COMP3297 plus common system-design practice.

Proof. Proof = N/A; Complete = pass the hardest design set.

Career paths

Required topics must reach **Sufficient**. Bonus topics strengthen a profile but are optional.

Quant Researcher

Required: QNT-205 Probability III, QNT-206 Statistical Inference, QNT-212 Regression II, QNT-216 Time Series II, QNT-301 Machine Learning, QNT-213 Research Methods, QNT-208 Bayesian, QNT-505 Portfolio & Asset Pricing II

Bonus: QNT-215 High-dimensional Stats, QNT-209 Causal Inference, QNT-211 Nonparametric, QNT-302 Deep Learning, QNT-204 Monte Carlo, QNT-507 Algo / Quant Trading, QNT-605 Probability Puzzles, QNT-704 LaTeX & Research Writing

Quant Trader

Required: QNT-202 Probability II, QNT-604 Game Theory, QNT-603 EV & Kelly, QNT-605 Probability Puzzles, QNT-503 Market Making, QNT-504 Options & Vol Trading, QNT-506 Microstructure II, QNT-602 Mental Math

Bonus: QNT-601 Fermi Estimation, QNT-401 Derivatives I, QNT-214 Time Series I, QNT-507 Algo / Quant Trading, QNT-102 Discrete Math & Combinatorics

Derivatives Quant

Required: QNT-407 Stochastic Calculus, QNT-408 Derivatives II, QNT-409 Interest Rate II, QNT-109 Numerical Methods II, QNT-204 Monte Carlo, QNT-410 Structured Products, QNT-411 Volatility Models

Bonus: QNT-402 Credit Risk, QNT-707 C++ II, QNT-205 Probability III, QNT-504 Options & Vol Trading

Risk Management

Required: QNT-404 Risk Management / VaR, QNT-402 Credit Risk, QNT-406 Operational Risk & Regulation, QNT-214 Time Series I, QNT-204 Monte Carlo, QNT-206 Statistical Inference, QNT-401 Derivatives I

Bonus: QNT-405 Risk Theory & Insurance, QNT-216 Time Series II, QNT-212 Regression II, QNT-403 Interest Rate I, QNT-408 Derivatives II

AI Researcher

Required: QNT-205 Probability III, QNT-110 Optimization, QNT-105 Linear Algebra II, QNT-302 Deep Learning, QNT-307 NLP, QNT-308 LLM / Generative AI, QNT-305 Reinforcement Learning, QNT-303 Learning Theory, QNT-111 Information Theory

Bonus: QNT-306 Computer Vision, QNT-208 Bayesian, QNT-215 High-dimensional Stats, QNT-704 LaTeX & Research Writing

ML Engineer

Required: QNT-301 Machine Learning, QNT-302 Deep Learning, QNT-705 Python, QNT-709 DSA II, QNT-710 Databases, QNT-712 Distributed Systems, QNT-304 MLOps, QNT-703 Dev Tools (Linux · Git)

Bonus: QNT-707 C++ II, QNT-308 LLM / Generative AI, QNT-714 Software Engineering & System Design, QNT-306 Computer Vision

Data Scientist

Required: QNT-206 Statistical Inference, QNT-212 Regression II, QNT-209 Causal Inference, QNT-301 Machine Learning, QNT-208 Bayesian, QNT-210 Experimental Design / A/B Testing, QNT-710 Databases

Bonus: QNT-214 Time Series I, QNT-211 Nonparametric, QNT-215 High-dimensional Stats, QNT-302 Deep Learning, QNT-703 Dev Tools (Linux · Git), QNT-704 LaTeX & Research Writing

Software Engineer

Required: QNT-709 DSA II, QNT-707 C++ II, QNT-711 Operating Systems, QNT-710 Databases, QNT-712 Distributed Systems, QNT-708 Computer Networks, QNT-714 Software Engineering & System Design, QNT-703 Dev Tools (Linux · Git), QNT-702 Computer Architecture

Bonus: QNT-713 Low-latency Programming, QNT-705 Python